Does Fisher–Rao Birth–Death Help Adaptive Biasing Force?
A Three-Case Study from a Model Potential to a Condensed-Phase Dimer

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Abstract

We study whether augmenting the Adaptive Biasing Force (ABF) method with a Fisher–Rao (FR) birth–death correction improves the computation of a reaction-coordinate free energy, and what kind of improvement it can reasonably be expected to provide. We test this on *three* systems that deliberately probe different regimes: a two-dimensional double-well *metastability* model with a quadrature reference; a two-dimensional *entropic-bottleneck* model with a narrow transverse channel, for which both the reference free energy and the conditional law $Y \mid X = x \sim \mathcal{N}(0, 1/(\beta\omega(x)^2))$ are analytic; and a many-body condensed-phase *Weeks–Chandler–Andersen (WCA) dimer* in a dense solvent, evaluated against a high-accuracy thermodynamic-integration reference. In each case the FR step reallocates walkers along the reaction coordinate by killing replicas in over-represented regions and cloning whole configurations from under-represented regions, nudging the empirical marginal $\widehat{p}_t$ toward a target q_t.

The central finding is not a single universal success story, but a regime map. In the metastability model, where plain ABF already samples reasonably well, estimated-target FR is a modest accelerator and the oracle diagnostic shows that target direction can matter. In the cold entropic bottleneck, ABF is starved and estimated-target FR gives a large gain (+51.3%, 20/20 seeds), but the uniform target nearly matches it and the oracle target is worse than ABF; here density correction and free-energy accuracy are not the same objective. A dedicated entropy-dominant stress test (transverse dimension m, total barrier held fixed) refines this further: as the barrier is made more entropic the *achievable* (oracle) gain grows specifically with the entropic share, yet the *deployable* estimated target does not capture it — an entropic-specific benefit that is real but target-limited. In the WCA dimer, tuned FR again improves the free energy reliably (+55.0%, 10/10 seeds), while the best accuracy occurs near the edge of an accuracy–diversity tradeoff. Across the cases, FR is best understood as a marginal-reallocation mechanism whose value depends on how starved the ABF estimator is, how quickly cloned orthogonal degrees of freedom re-equilibrate, and how aggressively birth–death erodes ancestor diversity. The oracle target is therefore a diagnostic control, not a deployable method and not a universal upper bound.

1 Introduction

1.1 Free-energy computation along a reaction coordinate

A central task in computational statistical mechanics is to compute the free energy associated with a low-dimensional *reaction coordinate* of a high-dimensional system in thermal equilibrium [8]. For most of the development we take a configuration to be a point $q = (x, y) \in \mathbb{R}^2$[1] drawn from the canonical (Boltzmann–Gibbs) density

$$\pi(x, y) \ \propto \ e^{-\beta V(x,y)}, \tag{1}$$

where V is the potential energy and $\beta = 1/(k_\mathrm{B}T)$ is the inverse temperature. Fixing the reaction coordinate to the first coordinate, $\xi(x, y) = x$, the free energy $F(x)$ and the local mean force $F'(x)$ along ξ are the objects of interest. The difficulty is metastability: when V has well-separated basins along x, unbiased Langevin dynamics crosses the barrier rarely, so a direct estimate of F converges intolerably slowly.

1.2 Adaptive Biasing Force

The Adaptive Biasing Force (ABF) method [3, 4, 6] removes this metastability adaptively. It maintains a running estimate $\widehat{F}_t'(x)$ of the mean force and applies the antiderivative $B_t(x) = \int^x \widehat{F}_t'$ as a bias that depends only on x. As $B_t \to F$, the effective dynamics along x becomes diffusive and the reaction coordinate is explored freely, while — crucially — the bias does not change the conditional law of the orthogonal degrees of freedom given x. ABF is mathematically well understood, with long-time convergence guarantees [7], and is a natural baseline against which to measure any proposed enhancement.

1.3 Why add a Fisher–Rao correction?

ABF accelerates exploration through the *force*, but it does not directly control *where the particles are*. Early in a simulation the estimated bias is poor, so the empirical reaction-coordinate marginal $\widehat{p}_t(x)$ can remain concentrated in the initial basin, starving other values of x of the samples ABF needs to estimate $F'(x)$ there. A Fisher–Rao (FR) birth–death correction [9] addresses exactly this: interpreting the evolution of $\widehat{p}_t$ as a gradient flow of relative entropy in the Fisher–Rao geometry [1], one reallocates probability mass by *killing* particles in over-represented regions and *cloning* particles in under-represented regions, driving $\widehat{p}_t$ toward a chosen target marginal $q_t(x)$ without integrating any additional dynamics. Coupling such a correction to ABF is appealing because the two act on complementary objects (force versus density), but it is not automatically beneficial: a birth–death step resamples *whole* configurations (x, y), and if it distorts the conditional law $Y \mid X = x$ it will corrupt precisely the quantity ABF estimates.

1.4 Central question and contributions

This report asks how FR changes ABF across *three* controlled systems, not whether one slogan can describe all of them:

1. **When does marginal reallocation help ABF, and when can it hurt?** (Q1)
2. **What target and rate choices make FR useful in each regime?** (Q2–Q3)
3. **What limitations and failure modes appear when the system becomes more realistic?** (Q4)

[1] We write a configuration as $q = (x, y)$ in the two-dimensional cases; the many-body case of Section 6 uses $q \in \mathbb{R}^{2n}$ with a scalar collective variable $z(q)$ in place of x (Remark 1). The symbol $q_t(\cdot)$, always carrying a time subscript and a scalar argument, denotes instead the Fisher–Rao *target marginal* of Section 3; the two are never used in the same role.

The systems are (i) a two-dimensional double-well *metastability* model with a quadrature reference (Section 4); (ii) a two-dimensional *entropic-bottleneck* model with a narrow transverse channel, for which both the reference free energy *and* the conditional law $Y \mid X = x$ are analytic (Section 5); and (iii) a many-body *Weeks–Chandler–Andersen (WCA) dimer* in a dense solvent with a high-accuracy thermodynamic-integration reference (Section 6). Together they span three regimes: an easy baseline where target direction can matter, a cold analytic bottleneck where density correction and free-energy accuracy can come apart, and a many-body system where improved coverage must be balanced against loss of lineage diversity.

Our contributions are: (i) a consistent ABF–FR notation that covers both the analytic two-dimensional models and the many-body WCA collective variable (Sections 2 and 3); (ii) three matched-seed production studies that quantify both the practical estimated-target method and diagnostic uniform/oracle targets (Sections 4 to 6); and (iii) a cross-case synthesis (Section 7) that explains why the conclusions are intentionally different across examples. A recurring methodological point is the careful interpretation of an *oracle* target that uses the reference free energy: it is a diagnostic control, never a deployable method and not a universal upper bound. Its behaviour is itself part of the evidence: best in the easy metastability setting, approximately tied with the practical target in WCA, and worse than ABF in the cold entropic bottleneck.

2 Background: ABF for a reaction-coordinate free energy

2.1 Free energy and the unbiased marginal

With reaction coordinate $\xi(x, y) = x$, the unbiased reaction-coordinate marginal of the canonical density (1) is

$$p_{\mathrm{ref}}(x) \;=\; \frac{\displaystyle\int e^{-\beta V(x,y)}\,\mathrm{d}y}{\displaystyle\iint e^{-\beta V(x,y)}\,\mathrm{d}x\,\mathrm{d}y}, \tag{2}$$

and the free energy along ξ is, up to an additive constant,

$$F_{\mathrm{ref}}(x) \;=\; -\frac{1}{\beta}\,\log p_{\mathrm{ref}}(x) + C. \tag{3}$$

Because F is defined only up to the constant C, every free-energy profile in this report — both estimates and the reference — is *centred* (its mean over the interior evaluation window is subtracted) before any L^2 error is formed; otherwise a meaningless constant offset would dominate the comparison. The reference quantities are computed by a one-dimensional y-quadrature on a fine grid with a log-sum-exp shift for numerical stability, which is exact up to discretisation and provides the ground truth $F_{\mathrm{ref}}, F'_{\mathrm{ref}}, p_{\mathrm{ref}}$.

2.2 The mean force is the object ABF estimates

Differentiating (3) and using $\partial_x p_{\mathrm{ref}}$ gives the *mean force*

$$F'_{\mathrm{ref}}(x) \;=\; \frac{\displaystyle\int \partial_x V(x,y)\,e^{-\beta V(x,y)}\,\mathrm{d}y}{\displaystyle\int e^{-\beta V(x,y)}\,\mathrm{d}y} \;=\; \mathbb{E}_\pi[\,\partial_x V(X,Y)\,|\,X = x\,]\,. \tag{4}$$

Equation (4) is the cornerstone of ABF: the gradient of the free energy equals the *conditional expectation of the instantaneous force* $\partial_x V$ given the reaction coordinate. It is a local, sample-based quantity — no integral of V is required — so it can be estimated on the fly from the particles currently at $X \approx x$, and the free energy is then recovered by quadrature, $\widehat{F}(x) = \int^x \widehat{F}'$.

2.3 Biased dynamics

ABF simulates N overdamped Langevin replicas that share one running mean-force estimate $\widehat{F}_t'(x) \approx F_{\mathrm{ref}}'(x)$ with bias potential $B_t(x) = \int^x \widehat{F}_t'$. Each replica (X_t, Y_t) evolves as

$$\begin{aligned} \mathrm{d}X_t &= \left[-\partial_x V(X_t, Y_t) + \widehat{F}_t'(X_t)\right] \mathrm{d}t + \sqrt{2/\beta}\, \mathrm{d}W_t^x, \\ \mathrm{d}Y_t &= -\,\partial_y V(X_t, Y_t)\, \mathrm{d}t + \sqrt{2/\beta}\, \mathrm{d}W_t^y, \end{aligned} \tag{5}$$

with independent Brownian motions W^x, W^y. The added force $+\widehat{F}_t'(X_t)$ acts *only* in the x-direction; the y-dynamics is unbiased. When $\widehat{F}_t' \to F_{\mathrm{ref}}'$ the net drift along x becomes $-\partial_x(V - F)$, whose x-marginal equilibrium is flat, so the reaction coordinate diffuses freely across the barrier.

2.4 Why the bias does not spoil the mean-force estimate

A bias that depends only on x leaves the conditional law of Y given $X = x$ invariant. This is what makes ABF self-consistent.

Proposition 1 (Conditional invariance under an x-only bias)**.** *Let ν be the canonical measure $\nu \propto e^{-\beta V(x,y)}$ and let $\nu_B \propto e^{-\beta\,[V(x,y) - B(x)]}$ be the measure biased by any function B of x alone. Then for every x the conditionals coincide, $\nu_B(y \mid X = x) = \nu(y \mid X = x)$, and consequently*

$$\mathbb{E}_{\nu_B}[\partial_x V(X, Y) \mid X = x] \;=\; \mathbb{E}_{\nu}[\partial_x V(X, Y) \mid X = x] \;=\; F_{\mathrm{ref}}'(x). \tag{6}$$

Proof. At fixed $X = x$ the bias factor $e^{\beta B(x)}$ is a constant in y and cancels between numerator and normaliser of the conditional density:

$$\nu_B(y \mid X = x) = \frac{e^{-\beta[V(x,y) - B(x)]}}{\int e^{-\beta[V(x,y') - B(x)]}\, \mathrm{d}y'} = \frac{e^{-\beta V(x,y)}}{\int e^{-\beta V(x,y')}\, \mathrm{d}y'} = \nu(y \mid X = x).$$

Taking the conditional expectation of $\partial_x V$ under this common conditional gives (6), the last equality being (4). □

Proposition 1 is the reason ABF works: even though the biased ensemble (5) has a different x-marginal than the canonical one, the conditional samples $Y \mid X = x$ remain canonical, so the empirical conditional average of $\partial_x V$ is an unbiased target for $F_{\mathrm{ref}}'(x)$. Any modification of ABF must respect this property; in particular, a birth–death correction that resamples whole configurations (x, y) is only admissible if it too leaves $Y \mid X = x$ (approximately) intact — a hypothesis we test directly in each of the three case studies (Sections 4 to 6).

Remark 1 (Generalisation to many-body systems)*.* Two of our three systems are two-dimensional with $\xi = x$, but the WCA dimer of Section 6 is a many-body system: a configuration is the full vector $q \in \mathbb{R}^{2n}$ of all particle coordinates, and the reaction coordinate is a scalar *collective variable* $z(q)$ (a scaled bond length) rather than a single Cartesian component. Proposition 1 holds verbatim under the substitution $x \mapsto z(q)$ and $y \mapsto$ the orthogonal complement of z on the constant-z manifold (here the solvent coordinates and the transverse dimer modes): a bias $B(z)$ that depends on the collective variable alone is constant on each level set of z, so it cancels in the conditional density and leaves the law of all orthogonal degrees of freedom invariant. Consequently the mean force along z — which for a nonlinear $z(q)$ also carries a geometric term (Section 6) — is still estimated without bias. The only requirement the FR step must meet is the same one as in two dimensions: a clone must copy the *entire* configuration $q \in \mathbb{R}^{2n}$, so that the orthogonal coordinates travel with z and the conditional law is preserved.

2.5 The ABF mean-force estimator

In the kernel form used by the reference implementation, $\widehat{F}'$ is a Nadaraya–Watson running average of the instantaneous force,

$$\widehat{F}'_n(x) \;=\; \frac{\sum_{k\le n}\sum_i K_h(x - X_k^i)\,\partial_x V(X_k^i, Y_k^i)}{\sum_{k\le n}\sum_i K_h(x - X_k^i)}, \tag{7}$$

with a Gaussian kernel K_h of bandwidth h, accumulated over all replicas i and past steps k; the free-energy estimate is its pinned trapezoidal antiderivative, $\widehat{F}_n(x) = \int_0^x \widehat{F}'_n$, and we use $B_t \equiv \widehat{F}$ as the bias. The GPU backend used for the production study replaces the $O(n_{\mathrm{grid}}\,N)$ kernel sums by an $O(N + n_{\mathrm{grid}})$ *binned-smoothed* estimator: particles are scattered onto the x-grid and the count and force histograms are Gaussian-smoothed, so that $\widehat{F}'(x) = \mathrm{smooth}(\mathrm{force})/\mathrm{smooth}(\mathrm{count})$ is a Riemann-sum approximation of (7) that converges to it as the grid spacing vanishes. The two estimators are validated to agree to well under 1% on static tests (Section 4.1).

3 The Fisher–Rao marginal correction

The Fisher–Rao (FR) correction is a birth–death / particle-reallocation step that acts on the *reaction-coordinate marginal* of the ensemble. It is interleaved with the ABF dynamics (5) and is designed to drive the empirical marginal $\widehat{p}_t(x)$ toward a target marginal $q_t(x)$, without adding any drift to the (x, y) dynamics.

3.1 Empirical and target marginals

At step n, after the Langevin proposal of all N replicas, the empirical reaction-coordinate marginal is estimated by a (reflected, edge-corrected) kernel density / binned-smoothed estimate of bandwidth η,

$$\widehat{p}_n(x) \;=\; \frac{1}{N}\sum_{i=1}^{N} K_\eta\big(x - X_n^i\big), \tag{8}$$

renormalised on the x-grid. The target $q_n(x)$ is one of three choices (Section 3.6); all are nonnegative and normalised, $\int q_n = 1$.

3.2 Fisher–Rao score

For each particle the FR *score* is the pointwise log-ratio of the empirical to the target marginal, with its ensemble mean removed:

$$S_i \;=\; \log\frac{\widehat{p}_n(X_n^i) + \varepsilon}{q_n(X_n^i) + \varepsilon} \;-\; \underbrace{\int \widehat{p}_n(z)\,\log\frac{\widehat{p}_n(z)}{q_n(z)}\,\mathrm{d}z}_{=\,\mathrm{KL}(\widehat{p}_n \,\|\, q_n)}, \tag{9}$$

where $\varepsilon > 0$ is a small regulariser and the integral baseline equals the Kullback–Leibler divergence $\mathrm{KL}(\widehat{p}_n \| q_n)$ (a grid quadrature; it coincides with the Monte-Carlo baseline $\frac{1}{N}\sum_k \log[\widehat{p}_n(X^k)/q_n(X^k)]$ up to discretisation). Subtracting the baseline makes the score mean-zero, so the birth–death step below conserves the expected particle number. The score is clipped to $[-S_{\max}, S_{\max}]$ for stability. Its interpretation is direct:

- $S_i > 0$: X^i lies where $\widehat{p}_n > q_n$, an *over-represented* region (too many particles);
- $S_i < 0$: X^i lies where $\widehat{p}_n < q_n$, an *under-represented* region (too few particles).

3.3 Birth–death resampling

The FR step approximates the Fisher–Rao gradient flow of $\mathrm{KL}(\widehat{p}\|q)$ as a fixed-N particle reallocation. Over a sub-step of length Δt and at effective rate γ_n^{eff} (Section 3.5), each *over-represented* particle ($S_i > 0$) is marked for death with probability

$$\mathbb{P}(i \text{ dies}) \;=\; 1 - \exp\!\big(-\gamma_n^{\mathrm{eff}}\, S_i\, \Delta t\big), \tag{10}$$

and particles with $S_i < 0$ supply clone sources. The WCA and metastability backends implement this in the direct death-replacement form: killed slots are filled by clones sampled from the under-represented pool, with weights increasing with $|S_i|$. The entropic-bottleneck backend uses a vectorised but closely related fixed-N survivor/clone-pool variant. This distinction matters for exact diagnostics, but not for the mechanism studied here: all variants move mass along the reaction coordinate by copying whole configurations from under-represented regions into over-represented ones.

Cloning copies the *whole* configuration (X^i, Y^i) — the score depends only on x, but the y-coordinate must travel with its x-coordinate, or Proposition 1 would be violated. The step is made deliberately gentle by three safeguards: (i) a burn-in and soft ramp on the rate; (ii) score clipping to $[-S_{\max}, S_{\max}]$; and (iii) a hard cap $f_{\max}$ on the fraction of particles replaced per event (`max_event_fraction`). The numerical values of $S_{\max}$, $f_{\max}$ and the FR stride `fr_every` are case-specific and are listed in each case's design table.

3.4 Notation across cases

The same FR principle is applied to all three systems, but the implementation and two parameter names differ slightly across backends. The birth–death *rate* is written γ in the analytic two-dimensional cases (Sections 4 and 5) and `fr_rate` in WCA (Section 6); they play the same role in (10). The *onset* of FR is parameterised as a burn-in *fraction* b in the analytic cases ((11), $n_{\mathrm{burnin}} = \lfloor b\, n_{\mathrm{steps}} \rfloor$) and as an absolute step count `fr_start_steps` in WCA; the two are related by $b \approx$ `fr_start_steps`$/n_{\mathrm{steps}}$. Where a case uses a collective variable $z(q)$ rather than x, read $x \mapsto z$ throughout this section (Remark 1).

3.5 Rate schedule

The FR rate is zero during an initial burn-in of $n_{\mathrm{burnin}} = \lfloor b\, n_{\mathrm{steps}} \rfloor$ steps (burn-in fraction b), then ramps softly to its maximum γ,

$$\gamma_n^{\mathrm{eff}} \;=\; \gamma\left(1 - e^{-(n - n_{\mathrm{burnin}})/n_{\mathrm{ramp}}}\right), \qquad n > n_{\mathrm{burnin}}, \tag{11}$$

and the birth–death step is applied every `fr_every` steps. The burn-in lets ABF build a usable mean-force estimate (hence a usable target) before any particles are moved; the ramp avoids a sudden shock to the ensemble.

3.6 Target marginals: estimated, uniform, oracle

The three targets differ only in which density q_n the score (9) is taken against. Recall $B_t = \widehat{F}_n$ is the current ABF bias.

Estimated target (the practical method). A target free energy $\widehat{F}_n^{\text{target}}$ is maintained independently of the bias by an exponential moving average of the ABF estimate, $\widehat{F}_{n+1}^{\text{target}} = (1-r)\,\widehat{F}_n^{\text{target}} + r\,\widehat{F}_{n+1}$, and the target marginal is the *biased* marginal that this target free energy would induce under the current bias,

$$q_n^{\text{est}}(x) \;\propto\; \exp\Big[-\beta\big(\widehat{F}_n^{\text{target}}(x) - B_t(x)\big)\Big]. \tag{12}$$

This uses only quantities available at run time and is the method whose practical value we assess.

Uniform target (a naive ablation).

$$q_n^{\text{unif}}(x) \;=\; \text{Unif}(x) \quad \text{on the } x\text{-grid}. \tag{13}$$

This tests whether simply *flattening* the reaction-coordinate marginal is enough. It is a target ablation, not the intended method; it ignores the fact that, under a bias $B_t \neq F$, the canonical biased marginal is not uniform.

Oracle target (a diagnostic control only).

$$q_n^{\text{oracle}}(x) \;\propto\; \exp\Big[-\beta\big(F_{\text{ref}}(x) - B_t(x)\big)\Big]. \tag{14}$$

Remark 2 (The oracle target is not an algorithm). Equation (14) replaces the estimated target free energy by the *reference* free energy F_{ref} — exactly the unknown quantity the whole procedure is trying to compute. The oracle target therefore *cannot* be evaluated in any real free-energy calculation. We include it solely as an ideal-information diagnostic: it asks what the FR birth–death step would do if the target marginal were anchored to the reference. This is not a universal upper bound. If oracle FR improves on estimated FR in a given regime, the gap can be read as target-shape headroom; if it ties or loses, the diagnostic instead says that the limiting mechanism is not the online target estimate, and may even be a mismatch between a fixed reference-shaped target and the transient ABF bias. The practical comparison is always

$$\text{ABF only} \qquad \text{versus} \qquad \text{ABF} + \text{estimated-target FR}.$$

3.7 The coupled ABF–FR step

One outer iteration, given the state $\{(X_n^i, Y_n^i)\}$ and accumulators, is:

1. update the ABF accumulators from the current particles and recompute $\widehat{F}'_n, \widehat{F}_n = B_t$ (Section 2.5);

2. propose the biased Langevin move (5) for all replicas;

3. if FR is active at this step (past burn-in, on the `fr_every` schedule): form $\widehat{p}_n$ (8) and the target q_n — one of (12), (13), (14) — compute the scores (9), and apply the birth–death step (10);

4. record diagnostics at the evaluation cadence.

The ABF estimator sees the post-resampling ensemble at the next step, which is the route by which improved x-coverage feeds back into a better mean-force estimate. The remainder of the report quantifies whether, and how, this feedback is beneficial.

4 Case I: a two-dimensional metastability model

Our first system is a two-dimensional double well: the regime in which ABF is already a strong baseline, so it sets the lower end of the difficulty spectrum.

4.1 Model potential and reference

We use a two-dimensional model potential with reaction coordinate $\xi(x, y) = x$,

$$\begin{aligned} V(x,y) = 3\,e^{-x^2}\Big(e^{-(y-1/3)^2} - e^{-(y-5/3)^2}\Big) \;-\; 5\,e^{-y^2}\Big(e^{-(x-1)^2} + e^{-(x+1)^2}\Big) \\ +\; 0.2\,x^4 + 0.2\,(y-1/3)^4. \end{aligned} \tag{15}$$

As shown in Figure 1, V has two metastable basins near $x = \pm 1$ separated by a barrier near $x = 0$ (with an upper sub-basin near $(0, 5/3)$), so unbiased dynamics is metastable in x. We fix $\beta = 4$. The reference $F_{\mathrm{ref}}, F'_{\mathrm{ref}}, p_{\mathrm{ref}}$ are obtained by the y-quadrature of (2)–(4) on a fine grid and are treated as ground truth.

4.2 Methods, metrics, and study design

Four methods are compared, identical except for the FR target: **ABF only** (no birth–death; baseline); **ABF+FR estimated** (12) (the practical method); **ABF+FR uniform** (13) (naive marginal-flattening ablation); and **ABF+FR oracle** (14) (diagnostic control, Remark 2). All four share the same potential, integrator, ABF estimator, seeds and initial conditions, so differences are attributable to the FR step and its target alone (matched-seed comparison).

4.2.1 Metrics

All profile errors are interior, domain-RMS L^2 distances on the window $x \in [-2.5, 2.5]$ (the reflecting walls and the x^4 confinement make the edges uninformative), with free-energy profiles centred as in Section 2.

Primary (practical) metrics.

$$\left\|\widehat{F}_T - F_{\mathrm{ref}}\right\|_{L^2}, \qquad \left\|\widehat{F}'_T - F'_{\mathrm{ref}}\right\|_{L^2}, \qquad \int_0^T \left\|\widehat{F}_t - F_{\mathrm{ref}}\right\|_{L^2} \mathrm{d}t. \tag{16}$$

The first two are *final-budget accuracy* at the end of the run; the *integrated* error rewards *fast* convergence (anytime performance), since a method that reaches a good profile earlier accumulates less error over $[0, T]$. These two notions need not be optimised by the same hyperparameters, a point we treat carefully in Section 4.3.

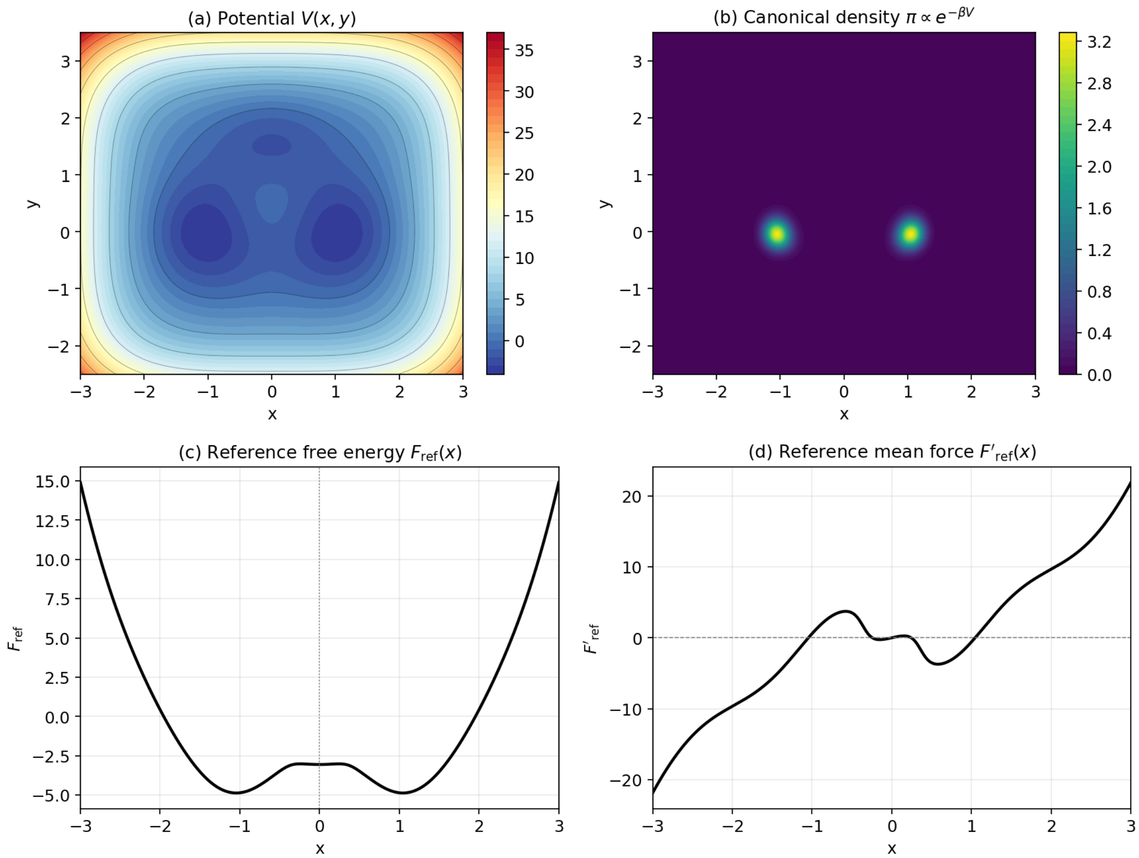

Figure 1: Problem geometry. (a) The potential $V(x, y)$ (15) with two basins near $x = \pm 1$. (b) The canonical density $\pi \propto e^{-\beta V}$ at $\beta = 4$, concentrated in the two basins. (c) The reference free energy $F_{\mathrm{ref}}(x)$ (a double well in x) and (d) the reference mean force $F'_{\mathrm{ref}}(x)$, both by y-quadrature.

Diagnostic metrics. Reaction-coordinate marginal flatness ($\|\widehat{p} - \mathrm{Unif}\|$ and $\|\widehat{p} - q\|$); cumulative barrier crossings at $x = 0$; region fractions (left basin / barrier / right basin); the FR event fraction (particles replaced per FR application) and the FR score dispersion $\mathrm{std}_i\, S_i$; conditional $Y \mid X = x$ errors (Section 4.2.1); and ancestor/diversity counts. We stress that *marginal flatness is a diagnostic, not the objective*: the goal is free-energy/mean-force accuracy (16). A method can flatten $\widehat{p}$ and yet fail to improve $\widehat{F}$ if it does so by corrupting the conditional samples.

Conditional $Y \mid X$ diagnostics. At a set of x-bin centres we compare the empirical conditional $\widehat{p}(y \mid x \in I_j)$ (the y-histogram of particles whose x falls in a narrow window) to the reference conditional $p_{\mathrm{ref}}(y \mid x) \propto e^{-\beta V(x,y)}$, via an L^2 distance and a KL divergence. By Proposition 1 a benign FR step should leave these close to the ABF-only values.

4.2.2 GPU implementation and study size

The production runs use a batched PyTorch backend that simulates many runs at once and reuses the reference metrics/diagnostics code verbatim. For speed it uses the binned-smoothed ABF/FR estimator of Section 2.5 rather than the exact kernel sums (7), and a fully vectorised fixed-N birth–death; the two backends agree to within the stated tolerance ($< 1\%$ in practice) on the validation suite. We flag in Section 8 that this binned estimator is a deliberate, but not mathematically

Table 1: Production study design (read from the result CSVs and `configs/two_dim_xi_x_production_gpu.yaml`).

Quantity	Value
Inverse temperature β	4.0
Time step Δt	0.002
Steps per run	100,000
Particles per run N	1000
Seeds	5 (0, 1, 2, 3, 4)
Methods	4 (ABF; FR estimated / uniform / oracle)
FR rate γ	0.01, 0.02, 0.05, 0.1
FR bandwidth η	0.075, 0.1, 0.15
FR burn-in fraction	0, 0.2, 0.4
FR stride `fr_every`	5
Total configurations	109
Total runs	545

Table 2: Hyperparameters of the selected configurations (integrated-$L^2(F)$ rule). n_{FR} is the Fisher–Rao stride `fr_every`.

Method	target	γ	η	burn-in	n_{FR}	seeds
ABF only	none	0	0.075	0	1	5
FR estimated	estimated	0.1	0.075	0	5	5
FR uniform	uniform	0.05	0.15	0.2	5	5
FR oracle*	oracle	0.1	0.075	0	5	5

identical, surrogate for the kernel notebook implementation. The FR methods sweep the rate γ, the marginal bandwidth η, and the burn-in fraction b at a fixed FR stride; the resulting study (Table 1) comprises 545 runs over 109 configurations and 5 seeds, at 1000 particles and 100,000 steps each, with *no* NaNs in the primary metrics. The selected configurations are listed in Table 2.

4.3 Results

Table 3 reports the four methods, each at its best configuration under the *integrated-$L^2(F)$* rule (convergence speed; see Section 4.3.2). Reading the practical comparison (ABF only versus ABF+FR estimated): the estimated-target FR reduces the median integrated free-energy error from 15.91 to 14.08, a 11.5% reduction, while the *final* free-energy error improves only slightly (from 0.0137 to 0.0126). The birth–death is extremely gentle: the median per-event replacement fraction is 1.6×10^{-5}, of order one particle in 10^5 per FR application. Figure 2 shows the same story over time: the FR curves detach below the ABF-only curve in the early and middle phase of the $L^2(F)$ panel.

4.3.1 Final profiles and target-type ablation

Figure 3 overlays the final mean-force, free-energy and reaction-coordinate-marginal profiles. All practical methods recover F'_{ref} and F_{ref} well; the visible differences between ABF only and ABF+FR estimated at the final budget are small, consistent with the modest final-error improvements. The x-marginal (panel c) is somewhat flatter under FR, as intended, but marginal flatness is a diagnostic, not the goal.

Table 3: Main comparison, configurations selected by smallest median *integrated* $L^2(F)$ (convergence speed). Lower is better except P(beats ABF).

Method	$\int_0^T \|\hat{F}_t - F_{\mathrm{ref}}\|\, dt$	$\|\hat{F}_T - F_{\mathrm{ref}}\|$	$\|\hat{F}'_T - F'_{\mathrm{ref}}\|$	P(beats ABF)	FR event frac.
ABF only	15.91	0.0137	0.0718	–	0
FR estimated	14.08	0.0126	0.0771	0.60	1.6×10^{-5}
FR uniform	14.46	0.0125	0.0769	0.80	5.3×10^{-6}
FR oracle*	13.27	0.0113	0.0602	1.00	1.7×10^{-5}

* The oracle target uses the reference free energy F_{ref} (the unknown being computed); it is a diagnostic positive control, *not* a usable algorithm. Per-config medians over 5 seeds; one configuration per method, selected by the smallest median integrated $L^2(F)$.

Table 4: Supplement: configurations selected by smallest median *final* $L^2(F)$ (final-budget accuracy).

Method	$\int_0^T \|\hat{F}_t - F_{\mathrm{ref}}\|\, dt$	$\|\hat{F}_T - F_{\mathrm{ref}}\|$	$\|\hat{F}'_T - F'_{\mathrm{ref}}\|$	P(beats ABF)	FR event frac.
ABF only	15.91	0.0137	0.0718	–	0
FR estimated	15.25	0.0120	0.0673	0.80	1.1×10^{-6}
FR uniform	14.97	0.0118	0.0680	0.60	7.3×10^{-6}
FR oracle*	13.27	0.0113	0.0602	1.00	1.7×10^{-5}

* Oracle is a diagnostic control (uses F_{ref}), not a deployable method. Configurations here are selected by smallest median *final* $L^2(F)$; cf. Table 3, which need not pick the same hyperparameters.

Figure 4 shows the distribution, over the 36 configurations of each target type, of the seed-median errors. For the *integrated* error all three FR target types have medians below the ABF-only baseline, but the spread is wide and some configurations are *worse* than ABF, so FR does not help unconditionally. For the *final* $L^2(F)$ the estimated and uniform medians sit essentially on the ABF baseline, and on the final *mean-force* error they lie slightly *above* it — only the oracle control is consistently below. Flattening the marginal with an imperfect target buys early speed but not final mean-force accuracy.

4.3.2 Two selection rules give different "best" hyperparameters

Because integrated and final errors measure different things, they need not be optimised by the same configuration. Selecting by *integrated* $L^2(F)$ (Table 3) favours fast convergence and selects an aggressive configuration for the estimated target ($\gamma = 0.1$, burn-in 0); selecting by *final* $L^2(F)$ (Table 4) favours final-budget accuracy and selects a gentle, delayed configuration ($\gamma = 0.01$, burn-in 0.4), which improves the final free-energy error by 12.5% and beats ABF on 80.0% of matched seeds. This dependence on the selection rule is itself a finding.

The *uniform* target is competitive here (integrated reduction 9.1%): under a near-converged bias $B_t \approx F$ the canonical biased marginal is already close to uniform, so "flatten $\widehat{p}$" and "match the biased marginal" nearly coincide. The *oracle* diagnostic (Remark 2) achieves the lowest errors of all (integrated reduction 16.6%), but it requires F_{ref} and is unusable; the correct reading is comparative: the estimated target captures 11.5% of the reduction against the oracle's 16.6%, so a substantial part of the mechanism's potential is left unrealised by the online target estimate. In this easy regime, then, the oracle–estimated gap localises the practical bottleneck in the *target estimation* — a reading we revisit, and qualify, in Section 7.

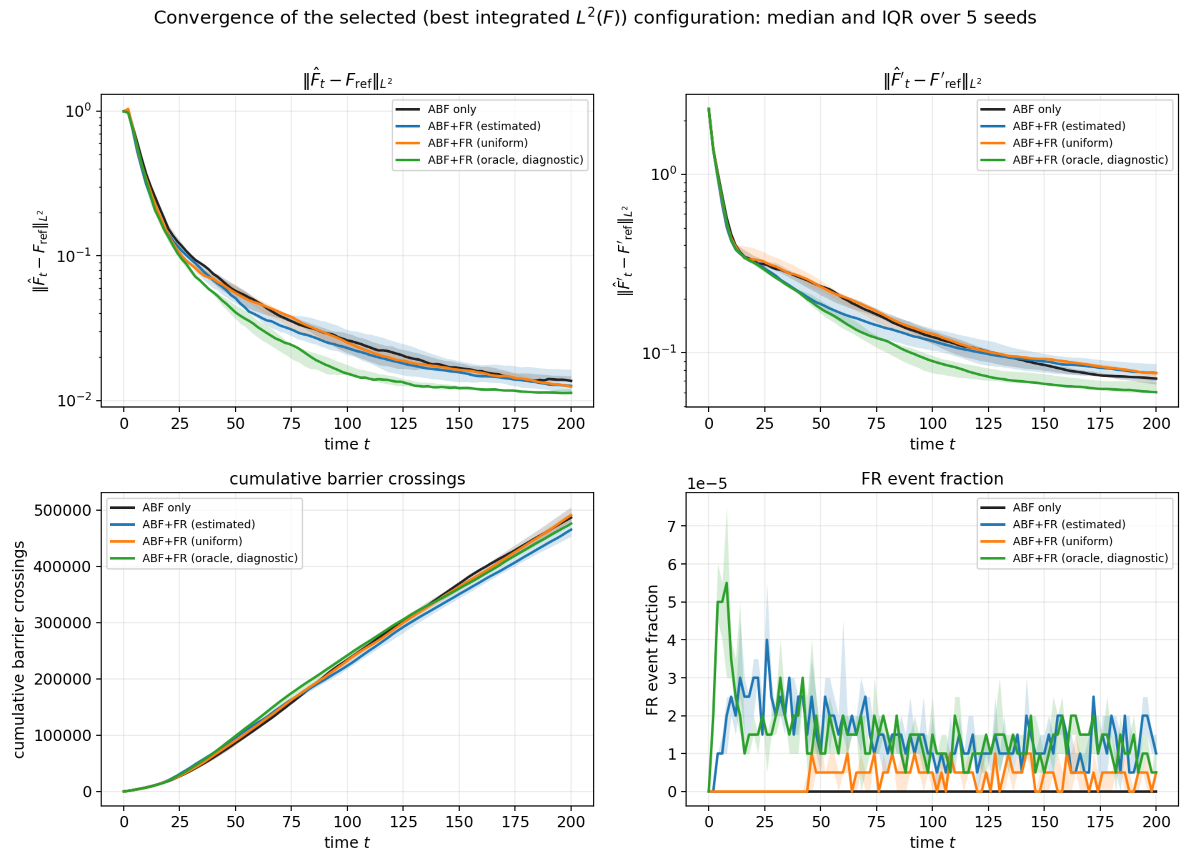

Figure 2: Convergence of the selected (integrated-$L^2(F)$ rule) configuration for each method: median (line) and inter-quartile range (band) over 5 seeds. The free-energy and mean-force errors (top, log scale) fall fastest for the oracle diagnostic and faster for the estimated/uniform FR than for ABF only in the early-to-middle phase; barrier crossings (bottom left) are essentially indistinguishable, and the FR event fraction (bottom right) is of order 10^{-5} throughout. The oracle curve is a diagnostic control (Remark 2).

4.4 Mechanism

The mechanism is a feedback loop between coverage and the mean-force estimate. ABF needs samples *across* the reaction coordinate: by (4), $F'_{\mathrm{ref}}(x)$ is a conditional average that can only be estimated where particles visit x. Early in a run the bias is poor and the ensemble is unbalanced — Figure 5(b) shows the initial occupancy split ($\sim$0.5 in one basin) relaxing only gradually toward the balanced 1/3–1/3–1/3 split. FR compares $\widehat{p}_t$ with the target q_t and preferentially kills over-represented and clones under-represented particles (10), raising the useful coverage of x earlier than diffusion alone would. Provided the birth–death does not corrupt $Y \mid X = x$ (Proposition 1), better coverage feeds back into a better conditional mean-force estimate.

A notable feature of Figure 5(c) is that FR does *not* produce many more barrier crossings than ABF; the benefit comes from *reallocating* existing replicas toward starved values of x rather than from manufacturing rare transitions. Consistent with this, the event fractions are of order 10^{-5}: in the successful regimes FR is a *weak* correction.

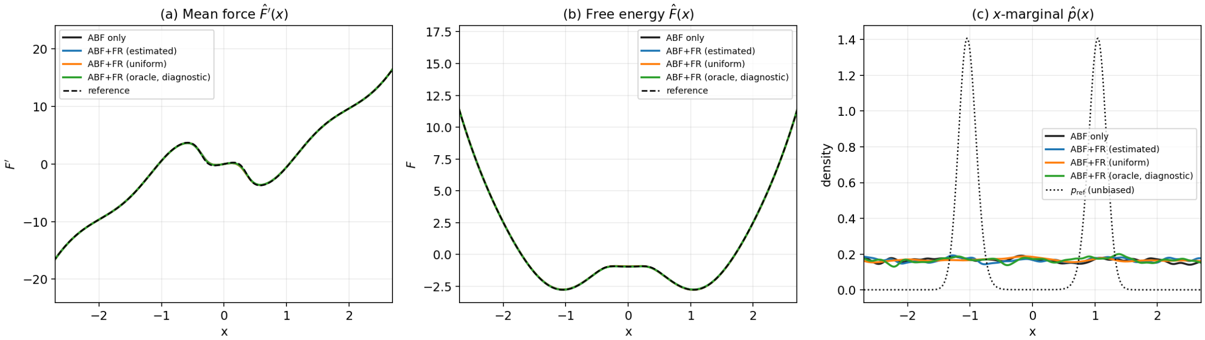

Figure 3: Final profiles (mean over seeds) for the selected configurations. (a) Mean force $\widehat{F}'(x)$ and (b) centred free energy $\widehat{F}(x)$ versus the reference (dashed); (c) reaction-coordinate marginal $\widehat{p}(x)$, with the unbiased p_{ref} (dotted). Practical methods recover the reference well; FR mildly flattens $\widehat{p}$.

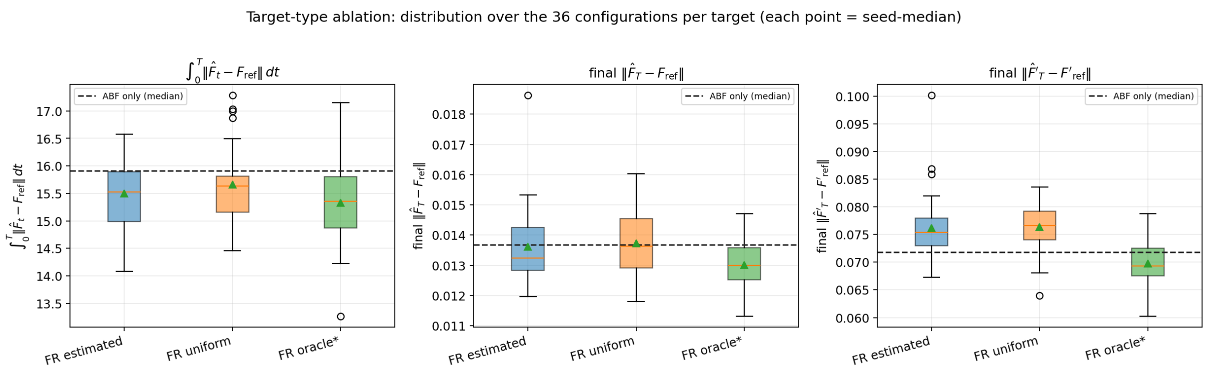

Figure 4: Target-type ablation: each box is the distribution over the 36 configurations of a target type (each point a seed-median); the dashed line is the ABF-only median. FR lowers the *integrated* error (left) but the practical targets barely change the *final* $L^2(F)$ (middle) and slightly raise the final mean-force error (right). Only the oracle diagnostic improves all three.

4.4.1 FR does not visibly corrupt $Y \mid X = x$

Figure 6 compares the conditional fidelity of all methods at the final time. The FR variants are *not* worse than ABF only; near the barrier ($x \in \{-0.5, 0, 0.5\}$), where ABF-only under-samples, they are in fact slightly *better* on both the L^2 and KL conditional metrics — direct evidence that the FR mechanism improves x-coverage without corrupting the orthogonal sampling, the benign behaviour Proposition 1 requires.

Remark 3 (Conditional diagnostics are not configuration-resolved). The metastability conditional-diagnostics output records only (method, target type, seed) per row, not the FR hyperparameters. It supports the *aggregate* statement that successful FR does not visibly corrupt $Y \mid X = x$, but cannot isolate a specific "good" versus "bad" configuration's effect. The entropic-bottleneck case (Section 5) closes this gap with an *analytic* conditional law.

4.4.2 What determines a good FR?

Figure 7 (estimated target) shows the integrated error decreasing as γ grows — a stronger correction converges faster, which is why the integrated rule selects the aggressive $\gamma = 0.1$ — but the same

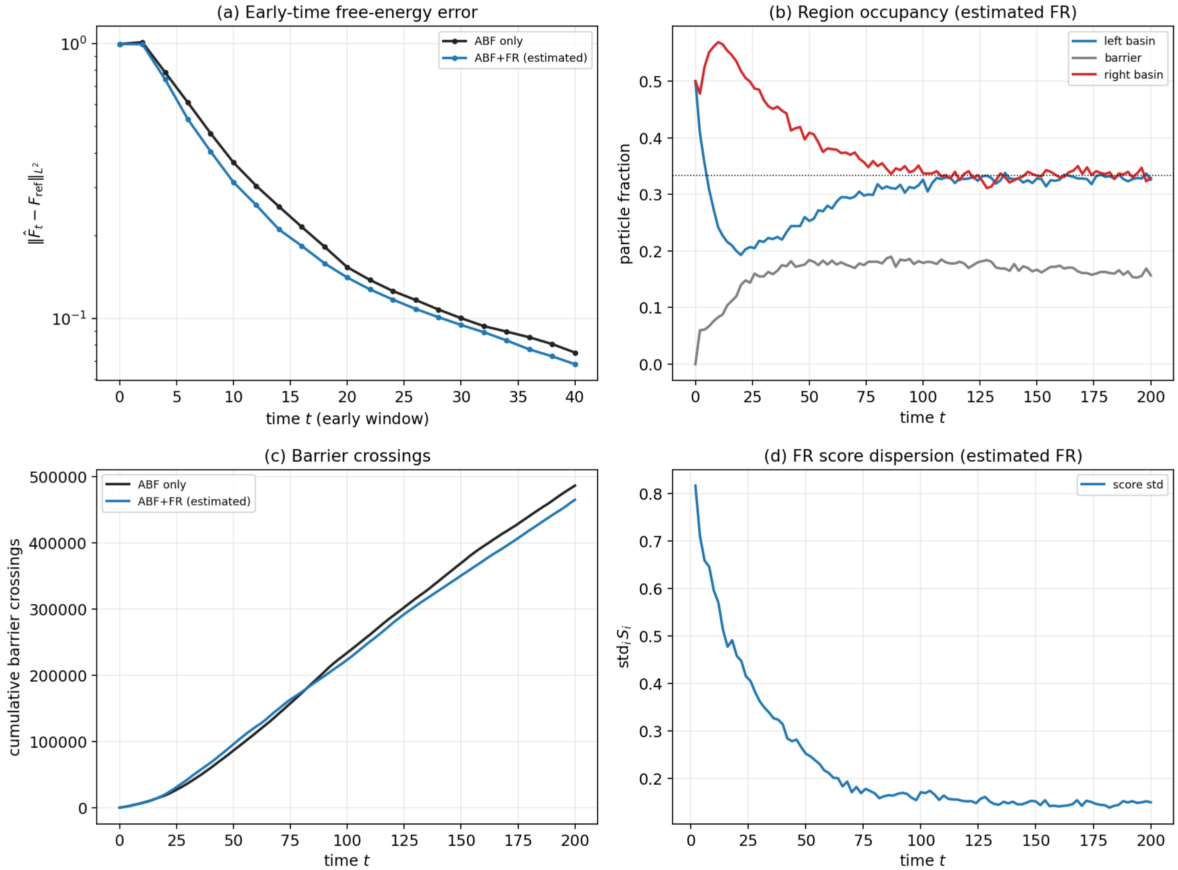

Figure 5: Mechanistic / early-time behaviour, estimated FR (best integrated-rule config) versus ABF only. (a) Early-time free-energy error (log scale): FR converges faster. (b) Region occupancy under FR relaxes from an unbalanced start toward the balanced 1/3 split (dotted). (c) Cumulative barrier crossings are nearly identical — the benefit here is redistribution, not a large increase in raw crossings. (d) FR score dispersion decays as $\widehat{p}_t \to q_t$.

setting does *not* minimise the final error (Section 4.3.2): there is an explicit speed-versus-accuracy trade-off in γ. The marginal bandwidth η is robust at intermediate values, and an early burn-in helps the integrated error here because the largest imbalance is at early times. Across the successful configurations the event fractions are tiny ($\sim 10^{-5}$) and the cap $f_{\max}$ never binds, so particle diversity is preserved.

4.5 Case finding

On this easy metastability example the answer to Q1 is *yes, but modestly, and chiefly as a convergence accelerator*. Practical estimated-target FR improves the integrated free-energy error by 11.5% with a smaller final-budget gain; the improvement is real but not a rescue, and it is not unconditional. Because plain ABF already converges well here, the room for FR to help is small. The oracle control does best, which — read *within this regime* — points to target estimation as the practical bottleneck; we will see in Sections 5 and 6 that this reading does not survive into the starved regimes, where the oracle's advantage vanishes or reverses.

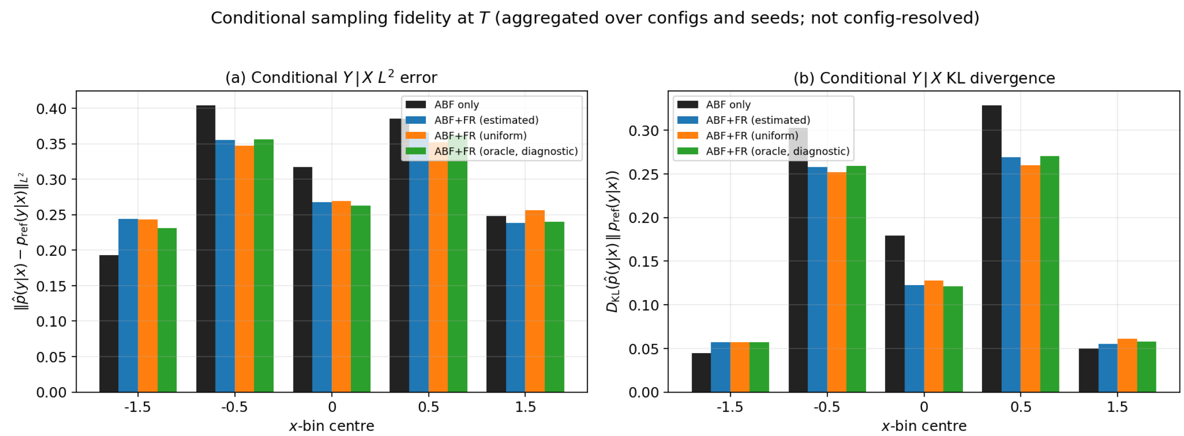

Figure 6: Conditional $Y \mid X = x$ fidelity at T, by x-bin: (a) L^2 distance and (b) KL divergence of $\widehat{p}(y \mid x)$ to $p_{\mathrm{ref}}(y \mid x)$. FR does not inflate the conditional error; near the barrier it slightly reduces it. *Limitation:* the conditional CSV identifies runs only by (method, target, seed) and is aggregated over configurations (Remark 3).

5 Case II: an entropic bottleneck

The second system tightens a transverse channel at the barrier top and, by lowering the temperature, drives ABF into a genuinely sample-starved regime. It also has a rare luxury: the conditional law $Y \mid X = x$ is *analytic*, so we can test Proposition 1 directly rather than against a histogram.

5.1 Model potential and reference

The potential couples a symmetric double well in x to a transverse harmonic channel whose stiffness varies with x,

$$V(x, y) = H\,(x^2 - 1)^2 + \tfrac{1}{2}\,\omega(x)^2\,y^2, \qquad \omega(x) = \omega_{\mathrm{out}} + (\omega_{\mathrm{in}} - \omega_{\mathrm{out}})\,e^{-x^2/2s^2}, \tag{17}$$

with reaction coordinate $\xi(x, y) = x$. The stiffness $\omega(x)$ is large near $x = 0$ (a *narrow channel at the barrier top*) and relaxes to ω_{out} in the wells. Because the y-integral is Gaussian, the reference is analytic up to a constant:

$$F_{\mathrm{ref}}(x) = H\,(x^2 - 1)^2 + \beta^{-1}\log\omega(x) + C, \qquad F'_{\mathrm{ref}}(x) = 4H\,x\,(x^2 - 1) + \beta^{-1}\,\omega'(x)/\omega(x), \tag{18}$$

and, crucially, the conditional law is the exact Gaussian

$$Y \mid X = x \;\sim\; \mathcal{N}\big(0,\; 1/(\beta\,\omega(x)^2)\big). \tag{19}$$

The base regime is $\beta = 8$, $H = 2.5$, $\omega_{\mathrm{out}} = 1$, $\omega_{\mathrm{in}} = 25$, $s = 0.25$, with $N = 256$ walkers, $\mathrm{d}t = 10^{-3}$, and 40 000 steps. The free energy is gauged at $x = 0$ and all L^2 errors are RMS over the interior window $x \in [-1.5, 1.5]$.

Why this is an entropic bottleneck — and an honest caveat. Near $x = 0$ the accessible y-width shrinks like $1/\omega(x)$: at the base setting the transverse channel is $\omega_{\mathrm{in}}/\omega_{\mathrm{out}} = 25$, i.e. its standard deviation is $25\times$ narrower than in the wells, contributing an *entropic* term $\beta^{-1}\log\omega(x)$ to the free energy. We are explicit, however, that in this parameterisation the entropic contribution

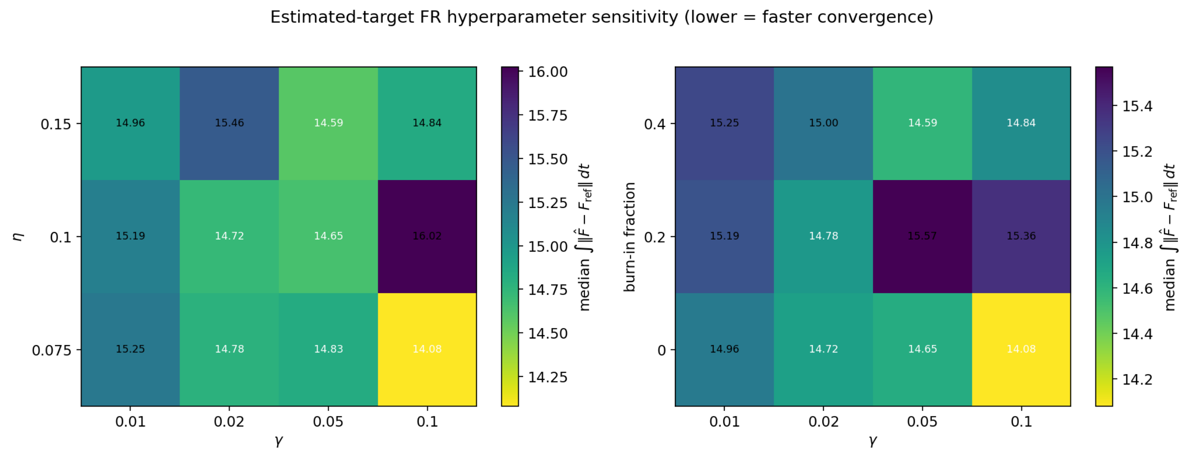

Figure 7: Estimated-target FR hyperparameter sensitivity: median integrated $L^2(F)$ over the grid (lower = faster convergence). Left: γ versus η; right: γ versus burn-in fraction. The integrated error improves with larger γ and earlier burn-in; intermediate η is robust. These trends should not be extrapolated beyond this regime.

Table 5: Entropic-bottleneck base regime and Fisher–Rao hyperparameters.

Quantity	Value
Inverse temperature β	8
Barrier height H	2.5
Channel stiffness $\omega_{\mathrm{out}}, \omega_{\mathrm{in}}$	1, 25
Channel width s	0.25
Walkers N	256
Time step Δt	10^{-3}
Steps per run	40,000
FR rate γ	15
FR stride `fr_every`	10
Score clip $S_{\max}$	3
Event cap $f_{\max}$	0.08
Target EMA rate	0.005

$\beta^{-1} \log(\omega_{\mathrm{in}}/\omega_{\mathrm{out}}) \approx 0.40$ is smaller than the energetic barrier height $U(0) - U(\pm 1) = H = 2.5$. The quantity $4H = 10$ is the quartic force prefactor in $F'(x)$, not the free-energy barrier. The bottleneck therefore modulates the transverse width but is *not* the dominant contributor to reaction-coordinate starvation; that role belongs to the energetic double well. This matters for reading the ω_{in} sweep below.

5.2 Methods and design

The practical comparison is again **ABF only** versus **ABF+FR estimated** (12), on matched seeds (identical initial conditions and Langevin noise per seed). The diagnostics **FR uniform** (13) and **FR oracle** (14) (built from the analytic F_{ref}, hence not deployable, Remark 2) are also run. The FR hyperparameters are listed in Table 5; an assertion guards every batched call so the reference never reaches the estimated target. For the ω_{in} sweep only, the stiffest channel is stabilised with a smaller time step and more steps at the same physical time horizon ($\mathrm{d}t = 4 \times 10^{-4}$, 100,000 steps).

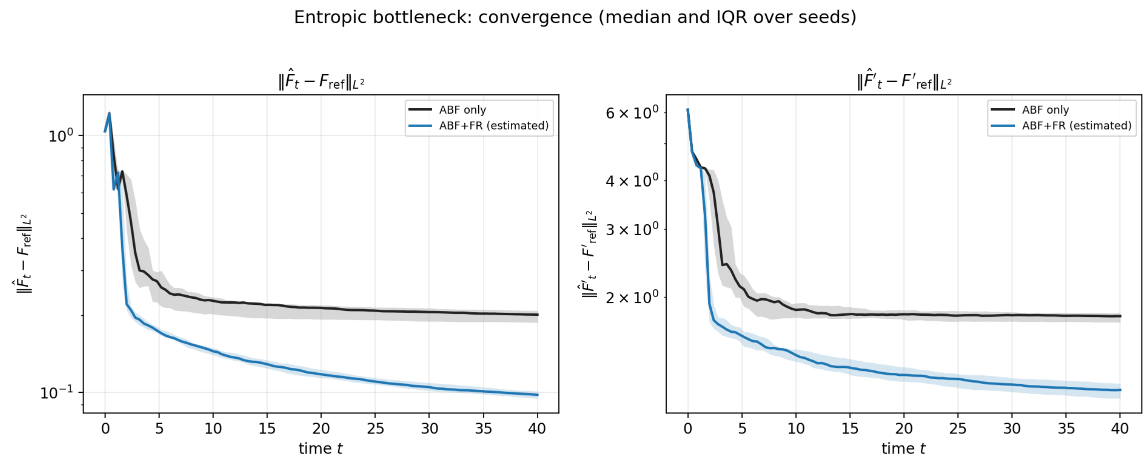

Figure 8: Entropic bottleneck, convergence (median and inter-quartile range over seeds). ABF+FR estimated separates from ABF early and stays below it for the entire trajectory on both the free-energy (left) and mean-force (right) errors.

Table 6: Temperature sweep at $\omega_{\mathrm{in}} = 25$: the FR gain flips sign as ABF crosses from well-sampled (hot) to starved (cold).

β	ABF $\|F - F_{\mathrm{ref}}\|$	FR $\|F - F_{\mathrm{ref}}\|$	gain (%)	win rate
2	0.055	0.060	-8.8	0.30
4	0.049	0.062	-26.1	0.00
8	0.201	0.098	+51.3	1.00
12	0.416	0.281	+32.3	1.00

5.3 Results

At the base regime the practical method reduces the median final free-energy error from 0.201 (ABF) to 0.098 (ABF+FR estimated), a 51.3% reduction, winning 20 of 20 matched seeds (Figure 8). Two diagnostics are immediately revealing. The *uniform* target — which carries no information about the free-energy shape at all — recovers almost the entire gain (50.5%), while the *oracle* target (the *analytic* F_{ref}) is *worse than plain ABF* (-28.1%). This is the opposite of the metastability ranking (Section 4.5): here a target that over-commits to a fixed shape fights the ABF bias instead of merely equalising coverage. The signature points to *balanced resampling / variance reduction across the reaction coordinate*, not free-energy-shape steering, as the source of the gain.

5.3.1 Temperature governs the gain, not the bottleneck width

Varying the inverse temperature β at fixed $\omega_{\mathrm{in}} = 25$ (Table 6) is the cleanest mechanistic signal in the study. FR *helps when sampling is hard and hurts when it is easy*: at high temperature ($\beta = 2, 4$) crossings are frequent, ABF already samples the whole coordinate (its error is ~ 0.05), and FR's birth–death merely injects resampling noise — it is mildly harmful and loses most seeds (-26.1% at $\beta = 4$). At low temperature ($\beta = 8, 12$) crossings are rare, ABF is starved on the far side of the barrier (error 0.2–0.4), and FR's balanced resampling substantially repairs the coverage, winning every seed. The cross-over sits between $\beta = 4$ and $\beta = 8$.

By contrast, sweeping the bottleneck width ω_{in} at fixed $\beta = 8$ (Figure 9) leaves the gain

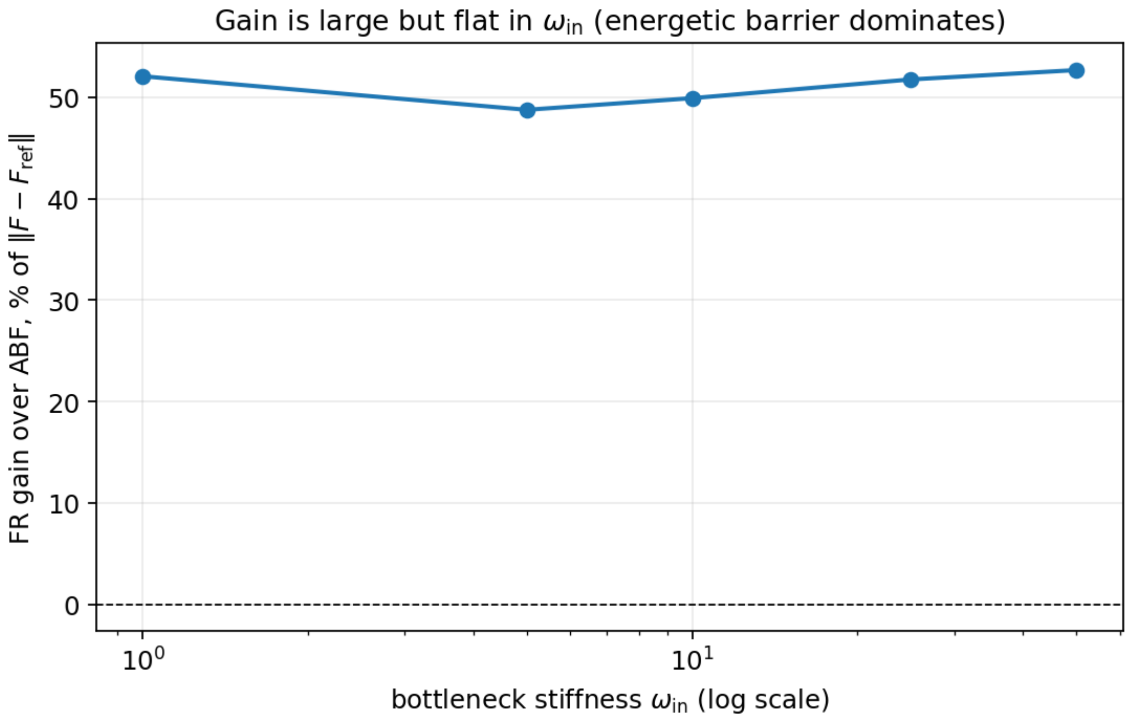

Figure 9: Bottleneck-width sweep ω_{in} at $\beta = 8$. The FR gain is large ($\sim 50\%$) and 100% reliable at every width, but it does *not* grow with the bottleneck strength: the ABF baseline error is flat because the energetic barrier height $H = 2.5$ dominates the smaller entropic contribution $\beta^{-1}\log(\omega_{\text{in}}/\omega_{\text{out}})$.

essentially *flat* at $\sim 50\%$. This is the honest consequence of the caveat above: the ABF baseline error is nearly independent of ω_{in} because the ω-independent energetic barrier dominates the sampling difficulty. The gain is large and reliable across ω_{in}, but it is governed by metastability (temperature), not by the bottleneck width. We do not claim a bottleneck-strength scaling the data do not show.

5.3.2 A wide safe regime for the FR rate

Sweeping the birth–death rate γ at the base regime (Figure 10), the gain increases *monotonically* up to $\gamma = 50$ (+72.6%); we did not reach a failure boundary within the swept range. The windowed ancestor effective sample size falls steadily from 101 to 9 as lineage diversity collapses, yet the free-energy error keeps improving. This differs from the WCA dimer (Section 6), where aggressive birth–death eventually degrades the estimator; the likely reason is the very short conditional-relaxation time of the analytic y-channel here, so a cloned walker re-equilibrates its orthogonal coordinate almost immediately. FR is unusually robust in this model.

5.4 Mechanism: conditional fidelity from an analytic law

Because $Y \mid X = x \sim \mathcal{N}(0, 1/(\beta\omega(x)^2))$ is *analytic* (19), we can test Proposition 1 directly rather than against a noisy histogram — the gap that Remark 3 left open in the metastability case. Figure 11 compares the empirical $\widehat{\mathrm{Var}}(Y \mid X \in I_j)$ to $1/(\beta\omega(x_j)^2)$ at five x-bins. The result is reassuring but not uniformly dominant: FR tracks the analytic variance well across nearly three orders of magnitude (from $\sim 2 \times 10^{-4}$ at the bottleneck $x = 0$ to ~ 0.12 in the wells), and is

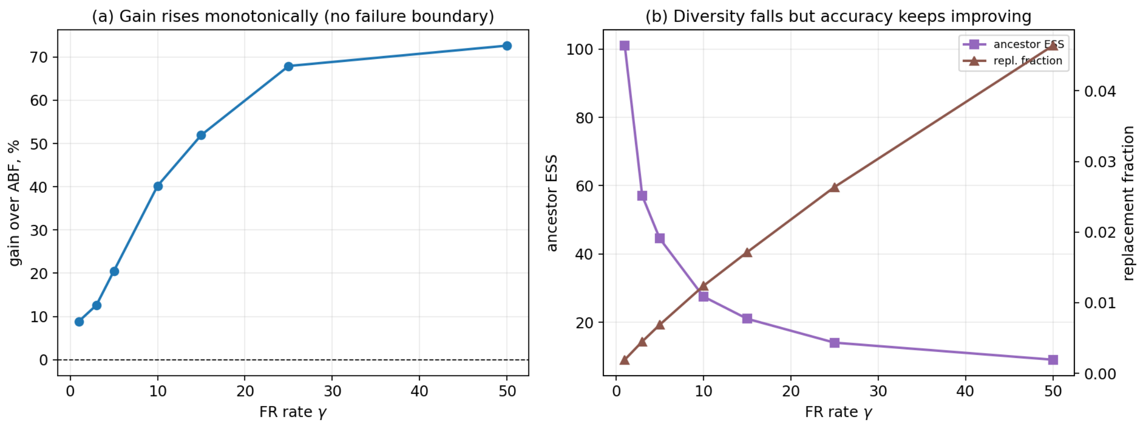

Figure 10: FR-rate sweep $\gamma \in \{1, \dots, 50\}$ at the base regime. Left: gain over ABF increases monotonically (no failure boundary in range). Right: windowed ancestor ESS collapses ($101 \to 9$) and the replacement fraction rises, yet accuracy keeps improving.

most competitive in the bottleneck and middle bins, but it is slightly worse than ABF in the well bins for this diagnostic. Thus the free-energy improvement is not coming from a visible collapse of the conditional law; it is achieved while the orthogonal variance remains close to the analytic conditional, with residual errors dominated by finite bin counts rather than a systematic FR bias.

5.5 An entropy-dominant stress test at fixed total barrier

The caveat of the previous subsections leaves one question open: the gain here tracks *temperature*, but because the energetic barrier dominates we never isolated whether an *entropic* barrier of the same height behaves any differently. We close that gap with a controlled stress test that promotes the transverse coordinate to m dimensions,

$$V(x, \mathbf{y}) = H\,(x^2 - 1)^2 + \tfrac{1}{2}\,\omega(x)^2\,\|\mathbf{y}\|^2, \quad \mathbf{y} \in \mathbb{R}^m, \qquad F_{\text{ref}}(x) = H\,(x^2 - 1)^2 + \tfrac{m}{\beta}\log\omega(x) + C, \tag{20}$$

so the marginal free energy carries an entropic term $\frac{m}{\beta}\log\omega(x)$ that we can enlarge at will (the conditional law generalises to $\mathbf{Y} \mid X = x \sim \mathcal{N}(0, I_m/(\beta\omega(x)^2))$). Writing the total thermal barrier as $B_0 = \beta H + m\log(\omega_{\text{in}}/\omega_{\text{out}})$ and the *entropic share* $\phi = m\log(\omega_{\text{in}}/\omega_{\text{out}})/B_0$, we set $H = (1 - \phi)B_0/\beta$ and $\omega_{\text{in}} = \omega_{\text{out}}\,e^{\phi B_0/m}$, which holds the total barrier *fixed* while sliding it from purely energetic ($\phi = 0$) to strongly entropic ($\phi = 0.9$). We sweep $\phi \in \{0, 0.25, 0.5, 0.75, 0.9\}$ at $\beta = 4$, $m = 2$, $B_0 = 8$ (20 matched seeds, time horizon $T = 40$); Table 7 lists the design and results.

The outcome is sharper than a simple yes/no, and is the most useful thing this case returns (Figure 12). The *deployable* estimated-target gain does *not* grow with the entropic share: it is U-shaped in ϕ (slope -8.3% per unit ϕ, correlation $r = -0.28$) and is mildly *harmful* in the middle of the sweep (-9.7% at $\phi = 0.5$), where the smooth entropic free-energy bump is hardest to estimate online from the high-variance instantaneous force $\omega(x)\omega'(x)\,\|\mathbf{y}\|^2$. The uniform target behaves almost identically, reaffirming that the practical mechanism is balanced resampling rather than free-energy-shape steering. The *oracle* target tells the opposite story: handed the analytic F_{ref}, its gain rises steeply with the entropic share (slope 64% per unit ϕ, $r = 0.94$), from harmful at $\phi = 0$ to $+43.7\%$ at $\phi = 0.75$. There is therefore a genuinely *entropic-specific* opportunity for birth–death — but it is *target-limited*: realising it needs a free-energy-shaped target accurate enough to steer against, which the online ABF-bias estimate is not in this regime. What the deployable method

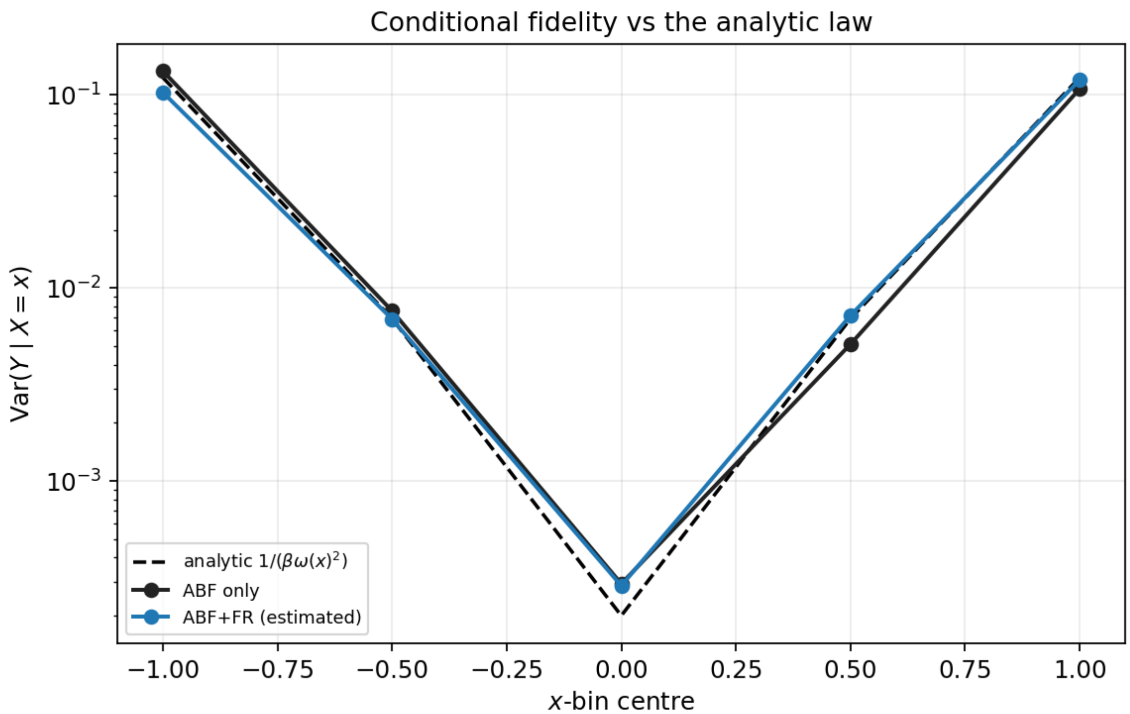

Figure 11: Conditional fidelity against the *analytic* law $\mathrm{Var}(Y \mid X = x) = 1/(\beta\omega(x)^2)$. Empirical FR and ABF variances both track the analytic curve across three orders of magnitude. FR is most competitive near the bottleneck and middle bins, while ABF is slightly closer in the well bins; the diagnostic shows no conditional-law collapse.

does deliver across the whole sweep is faster convergence: its integrated-error gain is positive at every ϕ (median +20.5%), and all FR variants preserve the m-dimensional conditional law as in the scalar case.

5.6 Case finding

The entropic bottleneck is a controlled bridge between the easy metastability case and the hard WCA dimer. Estimated-target FR robustly and substantially reduces the free-energy error in the cold, sample-starved regime ($\sim$ 51.3%, 20/20 seeds), but the gain is governed by *temperature* (FR helps when ABF is starved, mildly hurts when it is not), not by the bottleneck width. The mechanism is balanced resampling / variance reduction, not shape steering: the uniform target nearly matches the estimated one and the oracle target is *worse* than ABF — the reverse of the metastability ranking. The safe FR regime is wide (no failure boundary up to $\gamma = 50$), and the analytic conditional diagnostic shows that the gain is obtained without a systematic collapse of $Y \mid X = x$, even though FR is not uniformly better than ABF on every conditional-variance bin. Finally, the dedicated entropy-dominant stress test (Section 5.5) settles the bottleneck-versus-temperature question that the ω_{in} sweep could not: holding the *total* barrier fixed and growing only the entropic share, the deployable gain still does not increase (it is U-shaped in ϕ), whereas the oracle gain rises sharply to +43.7% at $\phi = 0.75$. The entropic-specific benefit is therefore real but *target-limited* — present for an exactly-shaped target, yet beyond the reach of the practical online estimator.

Table 7: Entropy-dominant stress test ($\beta = 4$, $m = 2$, $B_0 = 8$, 20 seeds, $T = 40$): at fixed total barrier, the deployable estimated/uniform gains are small and non-monotone in ϕ, while the oracle gain rises sharply — the entropic-specific headroom is real but target-limited.

ϕ	H	ω_{in}	ABF $\|F - F_{\text{ref}}\|$	est. (%)	est. wins	uni. (%)	oracle (%)
0	2.00	1.0	0.0464	+13.0	20/20	+13.0	-6.5
0.25	1.50	2.7	0.0360	+4.7	15/20	+6.0	-5.7
0.5	1.00	7.4	0.0286	-9.7	4/20	-14.3	+9.2
0.75	0.50	20.1	0.0248	-8.1	7/20	-0.5	+43.7
0.9	0.20	36.6	0.0312	+12.7	15/20	+12.9	+42.4

Total barrier $B_0 = \beta H + m \log(\omega_{\text{in}}/\omega_{\text{out}})$ ($= 8$) is held fixed while ϕ slides it from energetic to entropic ($H = (1-\phi)B_0/\beta$, $\omega_{\text{in}} = \omega_{\text{out}} e^{\phi B_0/m}$). Gains are matched-seed medians in final $\|F - F_{\text{ref}}\|$ vs ABF; the oracle target uses the analytic F_{ref} and is a diagnostic control, *not* deployable.

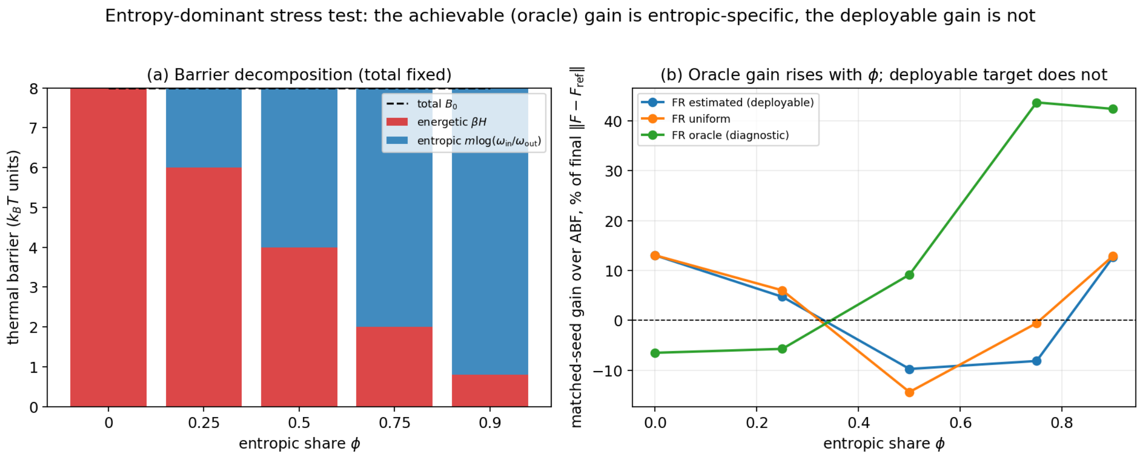

Figure 12: Entropy-dominant stress test at fixed total barrier B_0. (a) The barrier decomposition: as ϕ grows the energetic part βH is traded for the entropic part $m \log(\omega_{\text{in}}/\omega_{\text{out}})$ with the total held constant. (b) Matched-seed gain over ABF: the deployable estimated (and uniform) targets are small and non-monotone in ϕ, whereas the oracle gain rises sharply with the entropic share — the achievable benefit is entropic-specific, but the practical online estimate does not capture it.

6 Case III: a condensed-phase WCA dimer

The third system is many-body and is where ABF is most starved: a dimer that must push a dense solvent aside to change its bond length. It tests whether the two-dimensional findings survive in a setting with thousands of degrees of freedom and a nonlinear collective variable.

6.1 Model and reference

A dimer of two tagged particles sits in a periodic two-dimensional bath of $n_{\text{dim}}^2 - 2$ Weeks–Chandler–Andersen (WCA) solvent particles ($n_{\text{dim}} = 10$, lattice spacing a, box $L = n_{\text{dim}}\, a$). All particles interact through the purely repulsive WCA potential [10] (Lennard–Jones cut and shifted at its minimum $r_0 = 2^{1/6}\sigma$). The two dimer particles additionally feel a symmetric double-well bond $V_{\text{dim}}(r) = h\,(1 - ((r - r_0 - w)/w)^2)^2$ with barrier height $h = 2$ and width $w = 2$. The reaction

Table 8: WCA dimer base system and tuned Fisher–Rao hyperparameters.

Quantity	Value
Inverse temperature β	1
Time step Δt	0.002
Steps per run	250,000
Replicas N	1024
Lattice spacing a	1.5
Dimer barrier h, width w	2, 2
FR rate `fr_rate`	0.10
FR onset `fr_start_steps`	20,000
FR stride `fr_every`	5
Score clip $S_{\max}$	2.0
Event cap $f_{\max}$	0.02
Target EMA rate	0.005

coordinate is the scaled bond length

$$z(q) = \frac{|q_1 - q_2| - r_0}{2w}, \tag{21}$$

a nonlinear collective variable of the full configuration $q \in \mathbb{R}^{2n}$ (Remark 1): $z \approx 0$ is the *compact* state, $z \approx 1$ the *stretched* state, and $z \in [0.25, 0.75]$ the *transition* region. The free-energy barrier near $z = 0.5$ is partly entropic / crowding-driven, since the dimer must displace solvent to cross it. Dynamics are overdamped Langevin at $\beta = 1$, $\mathrm{d}t = 0.002$, with the minimum-image convention; the default system is $a = 1.5$, $\sigma = \epsilon = 1$. The local mean force along z is the constrained projection of the physical forces, including the geometric contribution with the sign convention of the estimator (Remark 1). The reference $F(z)$ is computed once per system a by constrained-MD thermodynamic integration [2, 5]; it is a high-accuracy numerical reference used for evaluation only, not an analytic exact solution. Errors are taken on the interior window $z \in [-0.1, 1.1]$ with the additive constant aligned.

6.2 Methods and design

As before the practical comparison is **ABF only** versus **ABF+FR estimated** (12), with **FR uniform** and **FR oracle** diagnostics; here the oracle target is built from the TI reference and is explicitly *not* deployable (Remark 2). A no-leakage assertion ensures only the oracle method ever receives the reference, and a clone copies the *entire* WCA configuration — not just z — so cloned solvent arrangements travel with the bond length. Replica lineage is tracked to report the ancestor effective sample size $\mathrm{ESS} = 1/\sum_a w_a^2$. The tuned FR configuration is given in Table 8; the rate is named `fr_rate` and the onset `fr_start_steps` (Section 3.4).

6.3 Results

The main comparison (Table 9; 10 seeds, $N = 1024$, 250 000 steps) shows the practical method **reliably beating ABF**: the tuned FR cuts the median final $L^2(F)$ from 0.0835 to 0.0361 (+55.0%), winning all 10/10 matched seeds. As in the entropic-bottleneck case, the uniform target (+54.6%) and the oracle target (+52.7%) are *essentially as good as* the estimated target — the online estimated target already extracts nearly all the available benefit, and there is no headroom to the

Table 9: WCA main comparison (250k steps, $N = 1024$, $a = 1.5$). The tuned row is the practical headline; the strong row shows the accuracy–diversity tradeoff. The mean-force gain is modest.

Method	$\|F - F_{\mathrm{ref}}\|$	$\|F' - F'_{\mathrm{ref}}\|$	gain (%)	wins	anc. ESS
ABF only	0.0835	0.3220	–	–	–
FR estimated (tuned)	0.0361	0.2513	+55.0	10/10	86
FR uniform	0.0372	0.2525	+54.6	10/10	90
FR oracle*	0.0387	0.2475	+52.7	10/10	100
FR estimated (strong)	0.0329	0.2590	+58.9	10/10	45
FR estimated (aggressive)	0.0935	0.3883	-13.1	1/10	22

* The oracle target uses the thermodynamic-integration reference F_{ref}; it is a diagnostic control, *not* a deployable method. Medians over 10 seeds.

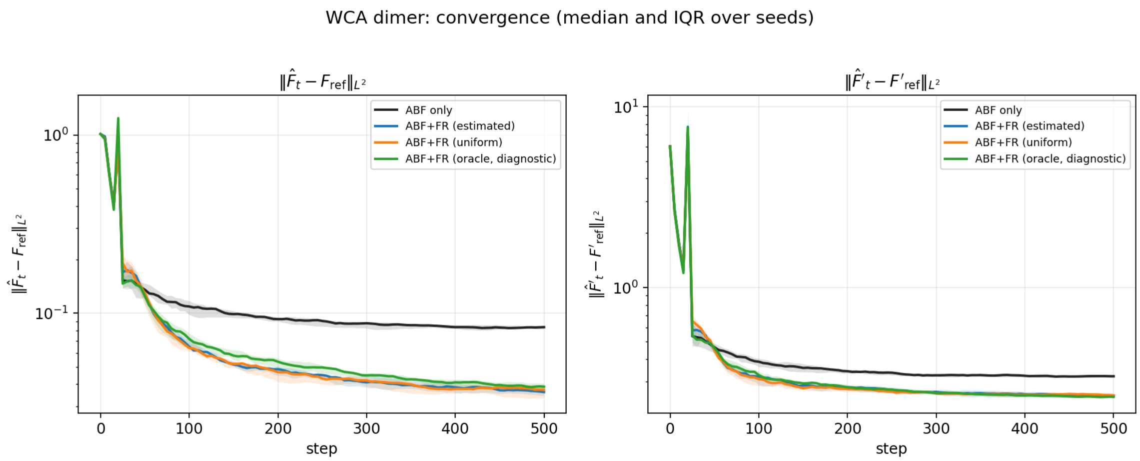

Figure 13: WCA dimer, convergence (median and inter-quartile range over seeds). ABF and FR are identical until FR activates; from then on FR is consistently below ABF on $L^2(F)$ for the rest of the run. The cumulative replacement count (right) confirms the birth–death is active but bounded.

oracle. The stronger estimated-target row in the same table attains the lowest final free-energy error, but with a much smaller ancestor ESS; we therefore use the tuned row as the practical headline and the strong row as the accuracy–diversity contrast. We are explicit that the headline gain is in the *free energy*: the mean-force error $L^2(F')$ improves only modestly ($\approx 22\%$), so we do not claim a large mean-force improvement.

6.4 Mechanism: support reallocation with a diversity cost

The diagnostics support the same broad mechanism as the entropic bottleneck rather than a stationary-coverage reshaping one. Final reaction-coordinate region fractions and the kernel-weighted ABF denominator in the transition window are nearly identical across ABF, FR estimated, FR uniform and FR oracle: FR does *not* converge to a different stationary z-distribution. Instead the gain appears immediately after FR activates and is sustained for the rest of the run. Figure 15 shows the interpretation: birth–death acts as a *support-reallocation resampler for the ABF estimator*. Replicas stuck in over-represented configurations are replaced by whole-configuration clones of under-represented ones, improving weighted force-sample support in important z-bins even when the marginal looks unchanged. A clone is not an independent solvent arrangement at birth; the point is that useful whole configurations are moved into starved reaction-coordinate regions and

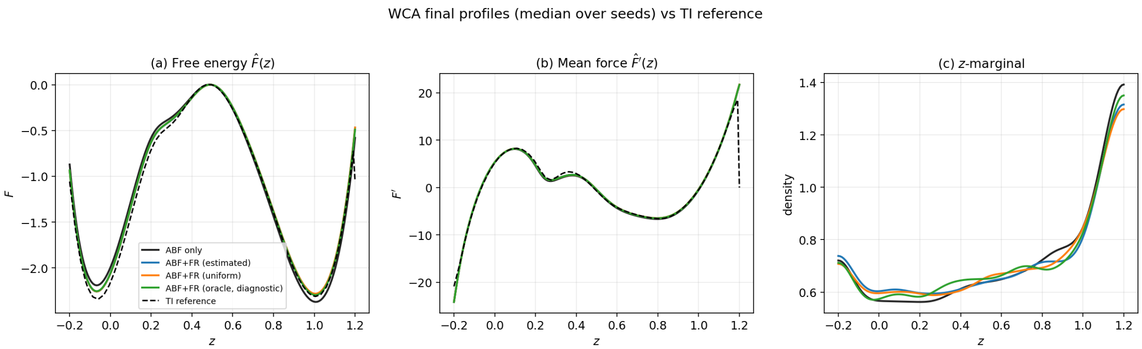

Figure 14: Final profiles (median over seeds) versus the TI reference (dashed): (a) free energy $\widehat{F}(z)$, (b) mean force $\widehat{F'}(z)$, (c) reaction-coordinate marginal. FR recovers the reference free energy more accurately than ABF, chiefly in and around the transition region.

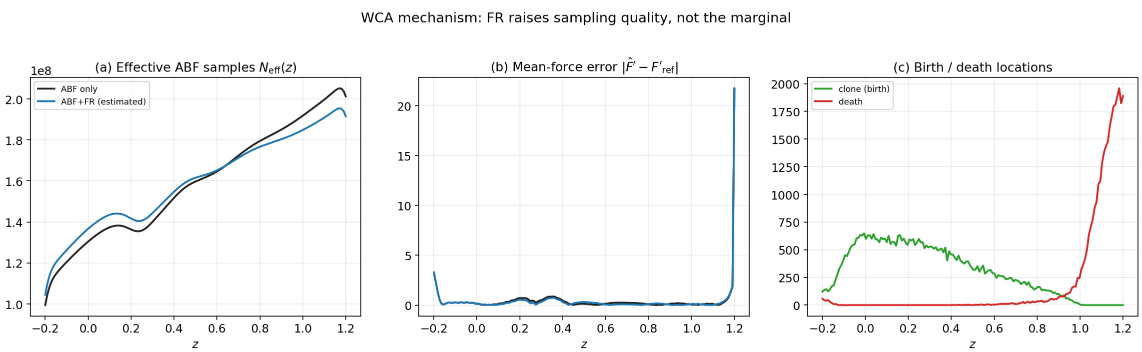

Figure 15: Mechanism. (a) The kernel-weighted ABF denominator $N_{\text{eff}}(z)$ is raised by FR in the transition window; (b) per-bin mean-force error is reduced there; (c) birth (clone) and death locations concentrate around the over/under-sampled regions; FR reshapes the *sampling support*, not the stationary marginal.

then decorrelated by subsequent Langevin dynamics. This explains why target shape matters little here, consistent with uniform ≈ estimated ≈ oracle.

6.5 Failure boundary

Unlike the entropic bottleneck, the WCA dimer exhibits a clear failure boundary in the FR rate (Figure 16). The rate ladder (a shorter 150k-step, 4-seed stage) maps a clean safe regime: too gentle (`fr_rate`≤ 0.05) gives a small gain (+19–29%); the sweet spot (`fr_rate`≈ 0.1–0.2) gives the maximal +58–64% with healthy ancestor ESS; the gain then *erodes* sharply (`fr_rate`$= 0.5$ falls to +12% with the error already climbing back toward the ABF baseline) and finally *reverses* for `fr_rate`≥ 1, where the median final $L^2(F)$ is worse than ABF (−38% at rate 1, −68% at rate 2) and no seed wins, as the ancestor ESS collapses from ~ 337 to ~ 20. The reversal arrives even earlier under a longer budget: the production aggressive configuration (`fr_rate`$= 0.5$ at 250k steps, 10 seeds) is already net-harmful (−13.1%, winning only 1/10 seeds, ancestor ESS ~ 22), because the extra steps give diversity loss more time to dominate. Over-aggressive birth–death destroys the replica diversity the mean-force estimator depends on, so the estimator degrades faster than coverage improves. This is the predicted failure mode; the analytic y-channel of the entropic

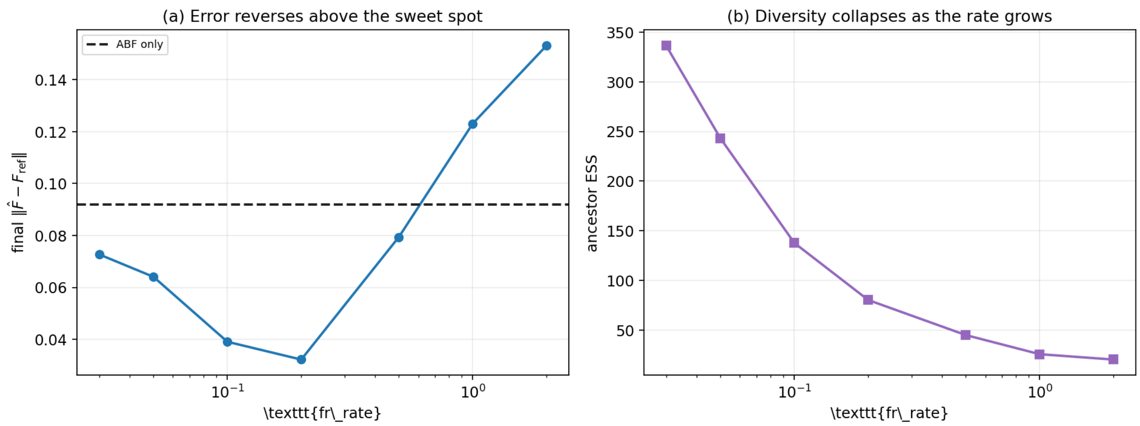

Figure 16: FR-rate ladder (150k steps, 4 seeds). Final $L^2(F)$ (left) is minimised in the 0.1–0.2 sweet spot, climbs back toward the ABF baseline (dashed) by `fr_rate`$= 0.5$, and rises above it for `fr_rate`≥ 1; ancestor ESS (right) collapses as the rate grows, pinpointing diversity loss as the failure mechanism.

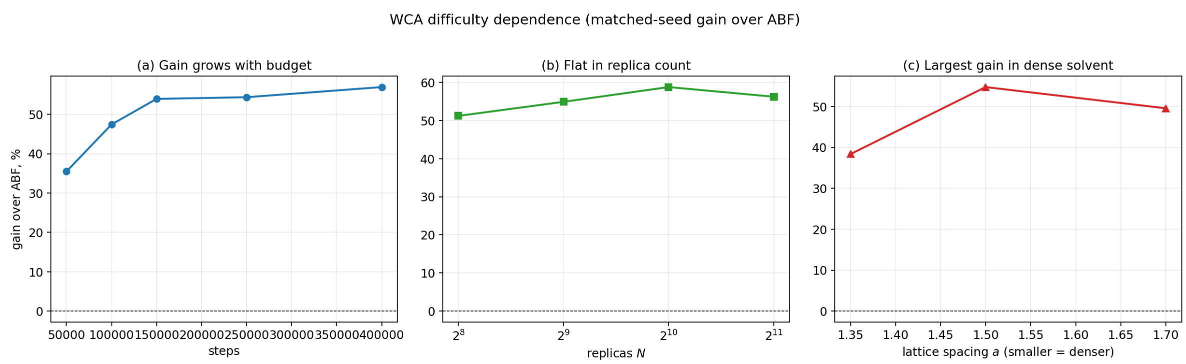

Figure 17: Difficulty dependence (matched-seed gain over ABF). (a) Budget: the relative gain grows with n_{steps} as ABF plateaus and FR keeps improving. (b) Replica count: gain is roughly flat. (c) Solvent crowding: smaller a (denser) yields the largest absolute error reduction. FR wins 5/5 seeds in every cell.

bottleneck masks it (Section 5) because cloned walkers re-equilibrate instantly, whereas a dense solvent does not.

6.6 Difficulty dependence

Varying one difficulty axis at a time (Figure 17) shows the gain is robust and, importantly, *grows with difficulty*. With *budget*, FR wins at every n_{steps} and the relative advantage grows rather than washes out (+35.5% at 50k to +56.9% at 400k): ABF's error plateaus around 0.08 after $\sim$ 150k while FR keeps descending, so FR is not merely an early-time accelerator that ABF later catches. With *replica count*, the gain is roughly flat (+51–59% from $N = 256$ to 2048) — FR helps even at $N = 256$. With *solvent crowding*, the densest bath ($a = 1.35$) makes the entropic barrier severe and ABF $\sim 4\times$ worse in absolute error; FR still wins 5/5 and delivers by far the largest absolute error reduction ($0.365 \to 0.225$). In the units a practitioner cares about — absolute free-energy error removed — the crowded regime is where FR helps most.

Table 10: Cross-case synthesis: FR is best read as regime-dependent marginal reallocation. The oracle target diagnoses target-direction headroom in the easy case, ties in WCA, and is harmful in the cold bottleneck. Metastability gain is on integrated error; the others use final $\|F - F_{\mathrm{ref}}\|$.

Case	Regime	FR gain (%)	wins	target effect
Metastability	ABF strong	+11.5 (int.)	–	oracle best
Entropic bottleneck (hot, β=4)	ABF easy	-26.1	0/10	n/a
Entropic bottleneck (cold, β=8)	ABF starved	+51.3	20/20	oracle worse
WCA dimer	ABF starved	+55.0	10/10	oracle $\approx$ est.

6.7 Case finding

On the condensed-phase WCA dimer, ABF+FR with an online estimated target reliably and substantially improves free-energy accuracy: +55.0% in median final $L^2(F)$ at the tuned operating point, winning 10/10 seeds, with the gain concentrated in the free energy (not the mean force). The mechanism is balanced-resampling variance reduction after warm-up; the precise target shape matters little (uniform $\approx$ estimated $\approx$ oracle, and the estimated target already reaches the oracle's accuracy). The method has a clear safe regime, `fr_rate`$\approx$ 0.1–0.2, and a real failure boundary: too-aggressive birth–death reverses the benefit as diversity collapses. The relative advantage *grows with simulation budget*, is insensitive to replica count, and removes the largest absolute error in the densest, hardest solvent — exactly the starved regime the support-reallocation mechanism predicts.

7 Cross-case synthesis

The three cases are most natural when read as three regimes of the same marginal reallocation idea, not as repetitions of one conclusion. FR always acts by moving ensemble mass along the reaction coordinate, but what that buys depends on the current bottleneck: target direction in the easy regime, variance/support in the starved regimes, and diversity control in the many-body regime.

7.1 A regime spectrum

At the easy end is the metastability model (Section 4), where ABF already converges well enough that the remaining headroom is partly about the *direction* of the target marginal. In the middle is the entropic bottleneck (Section 5), whose temperature sweep moves the same model from easy to starved and shows that density correction need not align with oracle-shaped free energy steering. At the hard end is the condensed-phase WCA dimer (Section 6), where the dimer must displace a dense solvent and the benefit is limited by the diversity of cloned many-body configurations. Table 10 lays the headline numbers side by side.

7.2 The gain follows starvation, but the mechanism changes

The practical, estimated-target FR gain tracks how starved the ABF estimator is. Where ABF is a strong baseline the gain is small and not unconditional — +11.5% on the integrated error in the metastability case, with a marginal final-budget gain — and where ABF is even less stressed (the hot entropic bottleneck, $\beta \leq 4$) FR is mildly *harmful* (-26.1% at $\beta = 4$), because birth–death injects resampling noise into an estimator that was not starved. Where ABF is starved the gain is large and reliable: +51.3% in the cold bottleneck (20/20 seeds) and +55.0% on the tuned WCA dimer (10/10 seeds). In WCA the relative gain grows with simulation budget, while the densest solvent has the largest absolute error reduction. The entropic-bottleneck temperature sweep is the cleanest evidence because it moves along the axis *within a single system*: the same model and code flip sign from $\beta = 4$ to $\beta = 8$ as ABF crosses from well-sampled to starved.

7.3 The oracle diagnostic is regime-dependent

The cases appear to disagree about the FR *target*. In the easy metastability case the oracle target is best and the estimated–oracle gap can be read, within that regime, as a target-quality bottleneck (Section 4.5). In the WCA dimer the oracle is no better than the estimated or uniform target, and in the entropic bottleneck the oracle is actually *worse than plain ABF*. This is not a contradiction; it is the point of the comparison.

When ABF is well sampled, the remaining error can be sensitive to the target *direction*, so knowing the reference shape helps. When ABF is starved, the first-order need is not an exact free-energy-shaped target but enough useful support across the reaction coordinate. A rough imbalance direction can then be sufficient, which explains why the uniform target nearly matches the estimated one in the two starved cases. A fixed oracle target can even fight the transient ABF bias in the cold bottleneck, so it loses to a neutral uniform target. The oracle is therefore a diagnostic of regime, not a performance upper bound.

The oracle has a second, more constructive use: it diagnoses *where* an exact free-energy shape would pay off, even when no practical estimate captures it. The entropy-dominant stress test (Section 5.5) makes this vivid. At fixed total barrier, increasing the entropic share flips the cold-bottleneck oracle from harmful to strongly beneficial (up to +43.7%), rising monotonically with the entropic fraction (slope 64% per unit ϕ), while the deployable estimated target captures essentially none of it (slope -8.3%). The achievable benefit is thus genuinely entropic-specific, but gated by the quality of the online free-energy target — the clearest single pointer this study gives for where a better estimated target would matter.

7.4 The shared failure mode is diversity collapse

Both the safe regimes and the observed failure are explained by ancestor diversity. Aggressive birth–death erodes the effective number of lineages; the gain survives only while the orthogonal degrees of freedom re-equilibrate fast enough to make cloned configurations useful. The WCA dimer reaches the failure boundary (`fr_rate`≥ 1 reverses the benefit in the rate ladder, and even `fr_rate`$= 0.5$ becomes harmful at production budget, ancestor ESS $\to$ 20–22) because its dense solvent re-equilibrates slowly. The entropic bottleneck does *not* reach a failure boundary even at $\gamma = 50$ (ESS $\to 9$) because its analytic y-channel re-equilibrates almost instantly. Same marginal-reallocation principle, different orthogonal relaxation times: a safe FR rate must be calibrated to the system, not copied across examples.

8 Discussion, limitations, and future directions

8.1 What the three cases establish (Q4)

The study establishes a regime map rather than a universal ranking. FR is a mild accelerator, and sometimes a source of extra noise, when ABF is already well-sampled. It becomes valuable when ABF is starved of reaction-coordinate support. In the many-body WCA case the same mechanism has a sharper operational limit: too much birth–death destroys ancestor diversity faster than the dense solvent can decorrelate cloned configurations. The different conclusions across examples are therefore not a defect of the study; they are the main information returned by the three-case design.

8.2 Limitations

We remain deliberately conservative about scope.

1. **The reaction coordinates are low-dimensional.** All three systems use a one-dimensional reaction coordinate (a Cartesian component in the 2D cases, a scalar bond length in WCA). Multi-dimensional or strongly curved collective variables, where the geometric mean-force term and the conditional sampling are far harder, are untested.

2. **The oracle target is not a method or an upper bound.** In every case the oracle uses the reference free energy, the unknown being computed, and is reported only as a diagnostic control (Remark 2). Its behaviour is read comparatively: best in the easy metastability regime, tied with practical targets in WCA, and worse than ABF in the cold entropic bottleneck.

3. **Production estimators are validated surrogates.** The 2D GPU backend uses a binned-smoothed mean-force/marginal estimator and a vectorised fixed-N birth–death variant; the WCA backend uses a constrained projection with the geometric term and a direct death-replacement step. These pass their validation suites and agree with the kernel/TI references to within tolerance, but are not bit-identical to the original kernel notebook implementation.

4. **Conditional diagnostics are incomplete in the many-body case.** The entropic-bottleneck case gives an analytic conditional check and shows no systematic collapse, but FR is not uniformly better than ABF in every bin. The WCA conditional fidelity is argued structurally (Remark 1) rather than measured per configuration.

5. **Entropic-specific gains are real but target-limited.** In the base Case II parameterisation the energetic double well dominates the difficulty and the ω_{in} sweep is honestly flat. The dedicated entropy-dominant stress test (Section 5.5), which holds the *total* barrier fixed and grows the entropic share, sharpens this: the *deployable* gain still does not scale with the entropic fraction (slope -8.3% per unit ϕ), whereas the *oracle* gain does (slope 64%, up to $+43.7\%$). The entropic-specific benefit exists but is gated by the quality of the online free-energy target; the practical estimator does not realise it here.

8.3 Failure modes

The analysis points to two related failure modes. First, an *overly aggressive mechanism* erodes ancestor diversity faster than the orthogonal coordinates can decorrelate clones. This is realised in the WCA dimer for `fr_rate`≥ 1 in the rate ladder, with even `fr_rate`$= 0.5$ turning harmful at production budget, and is masked in the entropic bottleneck because its analytic channel re-equilibrates rapidly. Second, a *misaligned target* can hurt when it over-commits to a fixed shape during a transient ABF bias. The metastability oracle–estimated gap quantifies target-direction headroom only in the easy regime; the entropic-bottleneck oracle shows that this reading does not generalise.

8.4 Future directions

- **Adaptive, diversity-aware FR strength.** Tie `fr_rate`/ γ to measured ancestor ESS or orthogonal-relaxation diagnostics, so the rate self-limits before diversity collapse.

- **Higher-dimensional and curved reaction coordinates**, where conditional sampling is genuinely hard and the geometric term is nontrivial.
- **Configuration-resolved conditional diagnostics** for the many-body case, to measure conditional fidelity rather than infer it from structure.
- **A better online target for entropic bottlenecks.** The entropy-dominant stress test (Section 5.5) shows the achievable (oracle) gain is entropic-specific but the ABF-bias estimate is too noisy to realise it. A lower-variance online free-energy estimator inside the FR target (e.g. a slower-EMA or eABF/CZAR-style estimate) is the natural way to convert that diagnosed headroom into a deployable gain.

9 Conclusion

Across three controlled systems — a 2D metastability model, a 2D entropic bottleneck with an analytic conditional law, and a many-body condensed-phase WCA dimer with a high-accuracy TI reference — Fisher–Rao birth–death helps ABF only after one specifies the regime. In the easy metastability setting it is a modest accelerator and the oracle diagnostic suggests target direction still matters. In the cold entropic bottleneck it gives a large practical gain (+51.3%, 20/20 seeds), but the oracle target is worse than ABF, showing that perfect reference-shaped density steering is not the same as best ABF accuracy. In the WCA dimer tuned FR reliably improves the free energy (+55.0%, 10/10 seeds), while stronger FR exposes the accuracy–diversity boundary.

The common object is marginal reallocation, not a universal variance-reduction slogan. FR moves whole configurations from over-represented to under-represented reaction-coordinate regions. When ABF is starved, this improves weighted support for the mean-force estimator; when ABF is already well sampled, it can add noise; and when the rate is too high, it collapses ancestor diversity. The oracle target is useful precisely because it reveals these regimes: it is best in the easy case, irrelevant in WCA, and harmful in the cold bottleneck. The practical recommendation is therefore regime-aware: use the estimated target, choose a moderate rate calibrated to orthogonal relaxation and ancestor diversity, and expect the largest returns where ABF alone lacks reaction-coordinate support.

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